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CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents the literature and studies relevant to the development of the proposed web-based recruitment platform for Fil International Recruitment Agency (FIRA), a Morocco-based, land-based recruitment agency that facilitates the overseas employment of Filipino workers. The platform integrates an Applicant Tracking System (ATS) with Sentence-BERT (SBERT) semantic matching to support centralized recruitment management while preserving human authority over final hiring decisions. The review examines the recruitment problem and its associated research gap, the web-based recruitment environment and the Philippine overseas employment regulatory framework, existing recruitment applications and their system features, the methodology and challenges of candidate-job matching, the algorithms and models underlying the proposed intelligent matching component, and the evaluation approaches used to assess software quality and matching accuracy.

The discussion is organized thematically across six major areas, with each theme bringing local and foreign sources together so that Philippine and international contexts can be compared directly. Local literature addresses the conditions of Philippine land-based recruitment agencies, the regulatory environment governing Overseas Filipino Worker (OFW) deployment, and locally developed matching systems. Foreign literature provides the established algorithmic foundations, current technologies, and global recruitment practices against which local conditions are measured. The reviewed literature and studies are analyzed and synthesized to determine existing practices,

identify limitations, and establish the research gap that the proposed FIRA platform seeks to address. Consistent with the guidelines for writing Chapter 2, the review does not merely summarize individual studies but analyzes, compares, and connects findings from multiple credible sources to build the theoretical and empirical foundation of the study.

PROBLEM

Recruitment management in the Philippine overseas employment context remains heavily mediated by manual and fragmented procedures that licensed agencies reconcile across separate tools and channels. Respicio and Co. (n.d.), in a study of land-based employment agencies that belong to the same category as FIRA, documented broker-mediated workflows in which agencies canvas, enlist, contract, and deploy workers through sequences that depend largely on paper records, spreadsheets, and private messaging. Such workflows persist in practice because they are familiar and inexpensive, yet they fragment applicant information across locations that are not simultaneously accessible to all authorized personnel. Recent reviews of electronic recruitment similarly observe that web-based methods have improved advertising and application capture, but that the integration of these stages with structured screening and decision support remains uneven, producing recurring gaps in recruitment efficiency research (Koman et al., 2024). Madanchian and Taherdoost (2026), in a bibliometric analysis of artificial intelligence in human resource management, found that although AI adoption in recruitment has grown rapidly, most implementations address isolated functions rather than the end-to-end recruitment pipeline.

The persistence of these gaps matters for FIRA because the agency's own environment, as documented in Chapter 1, exhibits precisely this fragmentation. Social media postings, online forms, spreadsheets, email, and local files each hold a partial view of the recruitment pipeline. Magdalenić and Luić (2025) emphasize that recruitment technology must be understood not as a collection of isolated tools but as an integrated ecosystem in which each stage feeds the next. When agencies treat job advertising, application capture, screening, and deployment documentation as separate concerns managed through different channels, the resulting information gaps propagate throughout the process. The literature collectively indicates that recruitment management has not yet been consolidated into a single, accessible, and auditable environment for smaller land-based agencies, which is the gap that a centralized platform must address.

Applicant tracking is a recognized weak point even where agencies have begun to adopt digital tools. Leony (2024) analyzed companies that implemented applicant tracking system software and confirmed that such systems streamline recruitment, while also identifying gaps in implementation quality, internal-user response, and measurable efficiency gains that justify continued research. The same study notes that adoption does not automatically translate into consistent pipeline visibility, because agencies often retain parallel manual records alongside the tracking system. Madanchian et al. (2024) reinforce this view, reporting that AI-enabled applicant tracking systems can lower recruiter cognitive burden and expand the applicant pool, but only when the system genuinely centralizes candidate data rather than duplicating it across tools.

In the Philippine OFW setting, Universal Staffing Services (n.d.) documents a deployment workflow consisting of sourcing, pre-screening, shortlisting, endorsement, documentation, and deployment that is conceptually linear but operationally scattered across documents and personnel, illustrating how tracking gaps emerge at each hand-off between agency and principal. Novaković et al. (2024) note that applicant tracking system adoption has transformed hiring, but that most platforms still treat the agency, the employer, and the applicant as separate, loosely coupled accounts rather than participants in a single end-to-end recruitment pipeline. Furthermore, Bevara et al. (2025), in developing Resume2Vec, observed that transforming applicant tracking systems with semantic representations requires rethinking how applicant data is structured and compared, not merely digitizing existing forms. These findings indicate that the absence of a single source of truth for application status, from New Application through FIRA Review, FIRA Approved, Employer Review, and Accepted or Declined, is a structural gap, not merely a tooling preference, and it is one that the proposed FIRA applicant tracking system is designed to close.

Matching applicants to job requirements is the recruitment stage where manual practice shows its clearest limitations. Aricheta, Santiano, and Ayo (2026), in developing HanapBuh.AI, explicitly frame the Philippine online job market as suffering from information overload and a heavy reliance on keyword-based search mechanisms that routinely fail to capture semantic skill alignment. Their observation is echoed internationally: Ajjam et al. (2025) demonstrated that semantic similarity matching outperforms keyword-based comparison on standard retrieval metrics, indicating that conventional keyword matching leaves substantial improvement unrealized. Moreover,

Kurek et al. (2024) found that an SBERT-family model surfaced the correct candidate within the top-500 results far more reliably than a keyword baseline. The gap, therefore, is not that matching is impossible but that existing tools remain anchored to lexical comparison, missing candidates whose qualifications are semantically equivalent but lexically distinct.

Existing recruitment platforms, although mature in feature breadth, do not fit the regulated agency-to-principal endorsement workflow that FIRA requires. Ajjam et al. (2025), in developing an AI-driven semantic matching framework, observe that most platforms treat matching as a candidate-side convenience rather than an agency-side decision-support function. Rosenberger et al. (2025) note that semantic matching research is rarely integrated with the documentation and verification stages that regulated cross-border recruitment demands. Novaković et al. (2024) note that most platforms treat the agency, the employer, and the applicant as separate accounts rather than participants in a single end-to-end recruitment pipeline. The reviewed literature points to an integration gap rather than a technology gap: the components exist, but they have not been combined into a single platform that serves the regulated, agency-mediated, cross-border context in which FIRA operates.

Among the studies reviewed, limited attention was given to the integration of regulated recruitment workflow, applicant tracking, and transparent semantic matching within a single platform designed for a Philippine land-based OFW recruitment agency. The reviewed literature provides the components, but no existing system combines them in a configuration that fits the FIRA context. This integration gap, rather than a technology gap, is the gap addressed by the proposed FIRA platform. The platform's

contribution is therefore architectural and contextual: it demonstrates how existing algorithmic components can be integrated into a regulated, agency-specific recruitment environment that preserves human authority over final hiring decisions while providing the centralized tracking, transparent matching, and verifiable documentation that the reviewed literature identifies as lacking.

Fil International Recruitment Agency (FIRA), the organization whose operations motivate the proposed platform, is a Morocco-based, land-based recruitment agency that facilitates the overseas employment of Filipino workers. The agency operates across two jurisdictions, coordinating its local agency in the Philippines, which handles applicant sourcing, screening, and endorsement, with its international principal office in Morocco, which handles employer relations and final selection coordination. FIRA's recruitment environment is shaped by the broader Philippine overseas employment context, in which the Philippine Statistics Authority (2025) estimates 2.19 million Overseas Filipino Workers deployed in 2024 and cash remittances reached approximately 262 billion pesos (Talabong, 2025), and by the bilateral framework within which the agency sources Filipino workers for Moroccan employers, a relationship now approaching fifty years of diplomatic recognition (Philippine News Agency, 2025). The detailed discussion of FIRA's stakeholders, recruitment workflow, and existing recruitment practices is presented in Chapter 2, where these elements are examined alongside the international and OFW recruitment framework within which the proposed platform operates.

The reviewed literature draws on both local and international sources to establish the recruitment context that the proposed FIRA platform addresses. Local sources

address Philippine conditions, local industry problems, local implementations, and local regulations. International sources provide the established algorithmic foundations, current technologies, and global recruitment practices against which local conditions are measured. The following subtopics organize the reviewed studies by category, allowing direct comparison between international and local findings.

Local studies on recruitment systems in the Philippines provide context for the operational realities that the proposed FIRA platform addresses. Enad and Pabalan (2023) studied the recruitment process in a Philippine land-based agency and documented the practices and challenges that characterize traditional workflows, including paper-based records, manual screening, and fragmented communication across stakeholders. Enad (2025) extended this work to examine applicant perceptions of fairness in Philippine overseas recruitment, finding that applicants value timely communication about application status as a primary indicator of a fair and respectful recruitment process. Mobo (2025) examined AI adoption in Philippine human resource management and found growing interest in AI-assisted recruitment, though implementation remains limited to larger organizations. Respicio and Co. (n.d.) and Cal International Philippines (n.d.) document the workflows and compliance requirements that Philippine land-based OFW recruitment agencies must satisfy, providing the local context against which the proposed platform's contribution is measured. Together, these local studies establish that the recruitment problems the proposed FIRA platform addresses are real, persistent, and shared across the Philippine land-based OFW recruitment sector.

International studies provide a richer evidence base on the design and effectiveness of web-based recruitment systems. Arman et al. (2023) conducted a PRISMA-guided review of electronic recruitment and found that web-based methods offer cost, time, and reach advantages over traditional methods. Lenasri and Shekhar (2025) performed a technology, classification, and effectiveness review identifying LinkedIn as the most dominant platform across sixty-two articles. Magdalenić and Luić (2025) applied design thinking principles to recruitment technology and argued that recruitment platforms must be understood as integrated ecosystems. Koman et al. (2024) and Madanchian and Taherdoost (2026) confirm that web-based recruitment has improved advertising and application capture but that integration with structured screening and decision support remains uneven. Bikou et al. (2024) model the recruitment process as a pipeline, while Santiago-Torner et al. (2026) found that process digitization improves outcomes most when it preserves the workflow's logical sequence. These international studies establish the design principles and evidence base against which the proposed FIRA platform's contribution is evaluated.

Studies on applicant tracking systems have examined their adoption, implementation quality, and impact on recruitment outcomes. Novaković et al. (2024) provide a three-class taxonomy of recruitment applications and note that most platforms treat the agency, the employer, and the applicant as separate accounts rather than participants in a single pipeline. Leony (2024) analyzed applicant tracking system implementation and identified gaps in implementation quality, internal-user response, and measurable efficiency gains. Madanchian et al. (2024) found that AI-enabled applicant tracking systems lower recruiter cognitive burden and expand the applicant

pool when the system genuinely centralizes candidate data. Hadžić et al. (2025) examined the integration of applicant tracking systems with broader human resource information systems and found that interoperability remains a challenge. Velankar and Khuspure (2025) surveyed one hundred human resource professionals and found that 86 percent reported that applicant tracking systems reduced time-to-hire, while 94 percent reported a positive impact on hiring outcomes. Bevara et al. (2025), in developing Resume2Vec, emphasized that transforming applicant tracking requires rethinking how applicant data is structured and compared, not merely digitizing existing forms.

Studies on AI-based recruitment have examined the application of machine learning to sourcing, screening, and matching. Madanchian and Taherdoost (2026), in a bibliometric analysis, found that AI adoption in recruitment has grown rapidly but that most implementations address isolated functions rather than the end-to-end recruitment pipeline. Ajjam et al. (2025) developed an AI-driven semantic similarity-based matching framework for recruitment. Rosenberger et al. (2025) describe the matching model as a trained artifact that captures the semantic relationships between applicant qualifications and job requirements. Spoladore et al. (2026) argue that hybrid algorithmic systems combining semantic representation with structured comparison consistently outperform single-signal approaches. Sony et al. (2025) examined AI governance in human resource management and argued that access control must be designed into the system architecture. Rigotti and Fosch-Villaronga (2024) emphasize the need for accountability and the right to challenge automated decisions in AI-based recruitment. Fabris et al. (2025) catalogued bias sources and fairness requirements in algorithmic

hiring. Lazaros et al. (2026) taxonomize human-in-the-loop patterns in AI-based decision systems. These studies establish the algorithmic and ethical foundations against which the proposed FIRA platform's matching component is evaluated.

Studies on semantic candidate matching have examined the application of contextual embedding approaches, particularly SBERT, to candidate-job comparison. Reimers and Gurevych (2019) introduced SBERT as a modification of BERT that produces semantically meaningful sentence embeddings suitable for efficient comparison through cosine similarity. Ajjam et al. (2025) demonstrated that semantic similarity matching outperforms keyword-based comparison on standard retrieval metrics. Kurek et al. (2024) found that an SBERT-family model surfaced the correct candidate within the top-500 results far more reliably than a keyword baseline. Rosenberger et al. (2025) demonstrate that domain-adapted SBERT embeddings improve matching accuracy over generic approaches. Rieon et al. (2026) report that SBERT combined with cosine similarity achieves strong retrieval performance in recruitment matching. Bevara et al. (2025) apply SBERT-based architectures to recruitment matching in their Resume2Vec work. Aricheta, Santiano, and Ayo (2026) developed HanapBuh.AI for the Philippine online job market, providing local context for semantic candidate matching. Abdelmotaleb, Sorkhabi, and Hassaan (2025) examined the impact of embedding choice on resume-job matching performance. Sutriawan et al. (2025) compared TF-IDF, Word2Vec, FastText, BERT, and GPT across matching tasks and found that contextual embedding approaches consistently outperform static and lexical approaches. These studies establish the algorithmic foundation that the proposed FIRA platform's matching component adopts.

Illegal recruitment and scam risks compound the problem. The Department of Migrant Workers (2026) reports the removal of tens of thousands of illegal recruitment posts from social media platforms, while Cabato and Magsambol (2025) document the escalation of illegal recruitment from 614 social media pages shut down in 2023 to over 70,000 fake online job offers thwarted in 2025. ABS-CBN News (2025) reports the closure of 32 illegal recruitment agencies in 2025 alone. The state policy established by Republic Act No. 8042, as amended by Republic Act No. 10022, treats syndicated illegal recruitment as economic sabotage, reflecting the severity with which the state views the exploitation of Filipino workers. The proposed FIRA platform contributes to the regulatory response by providing a verifiable, regulated alternative to the informal social-media-based channels where illegal recruitment flourishes.

The stakeholders affected by these gaps include recruitment agencies, foreign employers or principals, applicants or Overseas Filipino Workers, and government regulatory bodies. For recruitment agencies, the gaps manifest as operational inefficiency and compliance risk. For foreign employers or principals, the gaps manifest as delays in receiving qualified, verified candidates. For applicants or Overseas Filipino Workers, the gaps manifest as opaque application status and the risk of falling victim to illegal recruitment. For government regulatory bodies, the gaps manifest as the difficulty of monitoring and enforcing compliance when recruitment activities are scattered across informal channels. Each stakeholder category has distinct needs that the proposed FIRA platform must address through its role-based architecture and five-status workflow.

Time-to-hire and recruitment cost inefficiencies are the operational consequences of the gaps identified above. SHRM (2026) reports cost-per-hire averages of \$5,475 for non-executive positions, providing a benchmark against which the proposed platform's cost reduction can be measured. Dalain and Yamin (2025) found that web-based platforms reduce time-to-hire by enabling simultaneous processing of multiple applications. Pin (2026a, 2026b, 2026c) reports that AI-assisted recruitment tools can reduce time-to-hire by up to 82 percent, save approximately \$27,000 per hire, and improve hiring outcomes across multiple metrics. Tracker-RMS (2026) reports that 86 percent of recruiters say applicant tracking systems reduced time-to-hire, while 94 percent reported a positive impact on hiring. For FIRA, these benchmarks establish the expected benefits of the proposed platform: reduced time-to-hire through centralized tracking and intelligent matching, and reduced cost-per-hire through the automation of screening and endorsement tasks that currently consume significant recruiter time.

Volume overload from manual screening of high applicant numbers is a specific operational challenge that the proposed FIRA platform's matching component is designed to address. HeroHunt.ai (2025) and Skondras et al. (2025) document the stage-based process structure that contemporary web-based recruitment platforms adopt to handle high application volumes. Pin (2026b) reports a 4.7 percent applicant response rate across 635,000 tracked applications on Indeed, underscoring the challenge of achieving response in high-volume recruitment environments. Aricheta, Santiano, and Ayo (2026) frame the Philippine online job market as suffering from information overload that effective matching can address. The proposed FIRA platform addresses volume overload through the SBERT-plus-cosine-similarity-plus-LightGBM

pipeline, which produces ranked candidate lists that allow recruiters to focus their screening effort on the most promising candidates rather than reviewing every application manually. Furthermore, Liu et al. (2025) report that batched, GPU-accelerated SBERT embedding can process approximately 1,000 resumes per minute, establishing that the proposed platform's matching component can handle the volume of applicants that a Philippine land-based OFW recruitment agency typically processes.

WEB-BASED RECRUITMENT PLATFORM

Web-based recruitment refers to the use of internet-mediated platforms for posting job vacancies, capturing applications, screening candidates, coordinating selection, and managing recruitment records through a centralized digital environment. Furthermore, Arman et al. (2023), in a systematic review, characterize web-based recruitment as a set of practices that leverage internet technologies to expand reach, reduce cost, and improve the currency of applicant data relative to traditional print and in-person methods. Madanchian et al. (2024) describe modern web-based recruitment platforms as ecosystems that combine job distribution, applicant tracking, communication, and increasingly artificial intelligence-assisted screening and matching within a single digital environment. Magdalenić and Luić (2025) emphasize that these platforms must be understood not as collections of isolated tools but as integrated systems in which each component feeds the next.

The defining characteristics of web-based recruitment include centralized data storage, role-based access, real-time status updates, and the capacity for

multi-stakeholder coordination. Tursunbayeva et al. (2025) found that web-based human resource information systems are adopted primarily for centralized data management, regulatory compliance, and analytics capability. Lenasri and Shekhar (2025) identify the dominant platforms in the literature and note that their effectiveness depends on integration with structured screening workflows rather than isolated job advertising. For a cross-border agency like FIRA, web-based recruitment offers the additional advantage of enabling authorized personnel in different countries to access a shared, consistent view of applicant information, job requirements, and application status, a capability that fragmented manual and spreadsheet-based practices cannot provide.

The web-based recruitment process follows a sequence of stages from vacancy posting through candidate selection, with each stage generating structured data that feeds the next. Bikou et al. (2024) model the process as a pipeline in which job posting, application capture, screening, shortlisting, endorsement, and employer selection are interconnected stages rather than discrete events. Lenasri and Shekhar (2025) describe the process as one in which technology reduces friction at each hand-off between stakeholders. Santiago-Torner et al. (2026) found that process digitization improves recruitment outcomes most when it preserves the workflow's logical sequence rather than re-engineering it around the technology.

For the proposed FIRA platform, the web-based recruitment process maps directly onto the agency's existing workflow: employers submit job inquiries and requirements, the agency posts vacancies, applicants register and submit profiles and documents, agency staff verify and screen applicants, qualified applicants are endorsed to the

employer, the employer reviews and selects candidates, and the application status is tracked through each stage. The five application statuses, namely New Application, FIRA Review, FIRA Approved, Employer Review, and Accepted or Declined, represent the stages through which an application progresses. HeroHunt.ai (2025) and Skondras et al. (2025) confirm that this stage-based process structure is consistent with contemporary web-based recruitment practice, where the platform's value lies in maintaining a consistent, auditable record of each application's progress rather than in replacing the human judgment that operates at each stage.

Web-based recruitment offers several well-documented advantages over traditional methods. Arman et al. (2023) identified five core advantages: lower cost, shorter cycle time, wider reach, better-quality response, and more up-to-date applicant data. Dalain and Yamin (2025) found that web-based platforms reduce time-to-hire by enabling simultaneous processing of multiple applications and by making applicant information immediately accessible to all authorized stakeholders. Madanchian et al. (2024) reported that web-based recruitment systems lower recruiter cognitive burden and expand the applicant pool when the system genuinely centralizes candidate data. iHire (2024) found that organizations using web-based recruitment platforms report higher satisfaction with hiring outcomes than those relying on manual methods. For cross-border agencies, the advantages are compounded by the need for multi-location access, as centralized access for personnel in two countries eliminates the information fragmentation that characterizes spreadsheet and email-based coordination.

User interface and user experience are critical to the adoption and effectiveness of web-based recruitment platforms, particularly when the platform serves multiple user

roles with different needs and device contexts. Parpiev (2025) conducted an experimental comparison of responsive and non-responsive recruitment interfaces and found that responsive design significantly improves user satisfaction and task completion rates, particularly for mobile users. Magdalenić and Luić (2025) applied design thinking principles to recruitment technology and argued that the interface must be designed around the tasks and contexts of each user role rather than as a generic portal.

For the proposed FIRA platform, the user interface must accommodate three distinct roles: applicants, who may access the platform primarily through mobile devices; FIRA local agency staff, who work at desktop computers; and foreign employers or principals, who access the platform from abroad. This three-role problem requires responsive design that adapts to different screen sizes and input methods. Washington (2023) examined responsive web design and web accessibility in the context of recruitment platforms and found that mobile-first design is essential for applicant-facing interfaces, where smartphone access dominates. The proposed platform adopts Next.js with Tailwind CSS to support responsive, component-based interface development, with light and dark themes, English and Filipino language support, and adjustable text and display settings to accommodate diverse user needs. The Baymard Institute (n.d.) provides guidance on usability heuristics that inform the platform's interface design, emphasizing clarity, consistency, and feedback.

The applicant tracking system is the core functional component that enables centralized monitoring of applicants throughout the recruitment process. Madanchian et al. (2024) found that AI-enabled applicant tracking systems lower recruiter cognitive

burden and expand the applicant pool when the system genuinely centralizes candidate data. Novaković et al. (2024) note that applicant tracking system adoption has transformed hiring, though most platforms treat the agency, the employer, and the applicant as separate accounts rather than participants in a single pipeline. Hadžić et al. (2025) examined the integration of applicant tracking systems with broader human resource information systems and found that interoperability remains a challenge. Santiago-Torner et al. (2026) found that auditable application tracking is driven by regulatory compliance requirements. Velankar and Khuspure (2025) surveyed one hundred human resource professionals and found that 86 percent reported that applicant tracking systems reduced time-to-hire, while 94 percent reported a positive impact on hiring outcomes.

Applicant information management is the foundational function of an applicant tracking system, responsible for capturing, storing, organizing, and retrieving applicant data. Bevara et al. (2025), in developing Resume2Vec, emphasized that transforming applicant tracking requires rethinking how applicant data is structured and compared, not merely digitizing existing forms. Universal Staffing Services (n.d.) documents that applicant information in Philippine overseas recruitment typically includes personal data, educational background, employment history, skills and certifications, submitted documents, and application status. The proposed FIRA platform manages applicant information through structured profiles that aggregate resume content, submitted documents, qualification records, and application history within a single record accessible to authorized personnel. Cal International Philippines (n.d.) confirms that

applicant information management is the first function that agencies seek to digitize, as it underpins all subsequent screening, endorsement, and deployment activities.

Application status monitoring makes the position of each applicant in the recruitment pipeline visible to authorized stakeholders in real time. Greenhouse (2025) reports that candidate ghosting, where applicants disengage without notification, affects 50 percent of candidates, underscoring the need for transparent status communication. The proposed FIRA platform tracks application status through five stages: New Application, FIRA Review, FIRA Approved, Employer Review, and Accepted or Declined. Each status update is reflected in the applicant's account, allowing applicants to monitor their progress without needing to contact agency staff for updates. Santiago-Torner et al. (2026) found that status transparency improves applicant satisfaction and reduces the coordination overhead associated with status inquiries. Enad (2025) found that applicants value timely communication about application status as a primary indicator of a fair and respectful recruitment process.

Recruitment pipeline management extends status monitoring to the visualization and control of the entire recruitment funnel, from application through deployment. Bevara et al. (2025) found that pipeline visibility is a primary driver of applicant tracking system adoption, as it enables agencies to identify bottlenecks and allocate resources accordingly. The proposed FIRA platform provides pipeline management through a dashboard that displays the distribution of applications across statuses, the progress of individual applications, and the overall recruitment workload. Hadžić et al. (2025) examined pipeline integration with broader human resource systems and found that end-to-end visibility improves recruitment outcomes. For FIRA, the pipeline must

accommodate the agency-specific endorsement workflow, where applicants move from agency screening to employer review and final selection, with the option to remain available for other opportunities if not selected for a particular vacancy.

Applicant endorsement and employer review represent the hand-off between the agency and the principal, where verified applicants are presented to the employer for final selection. Cal International Philippines (n.d.) documents that endorsement requires the transmission of complete applicant information, including qualifications, documents, and screening results, to the employer for review. The proposed FIRA platform manages this hand-off through the Employer Review status, where endorsed candidates' profiles are presented to the employer with the results of the matching component as decision-support information. Greenhouse (2025) reports that structured endorsement, where candidates are presented with consistent, comparable information, improves employer decision quality. The platform's matching results, including percentage similarity scores and ranked candidate lists, are presented as advisory information that supports the employer's review without replacing human judgment about factors not captured in textual profiles.

Recruitment reporting converts the records that the applicant tracking system accumulates into actionable information for monitoring, evaluation, and compliance. Santiago-Torner et al. (2026) found that reporting capabilities are driven by regulatory compliance requirements, as agencies must demonstrate adherence to documentation and verification standards. The proposed FIRA platform provides reporting functions that summarize application volumes, status distributions, time-to-hire metrics, and recruitment outcomes. Bevara et al. (2025) note that reporting transforms applicant

tracking data from operational records into strategic information that supports continuous improvement. For a regulated agency like FIRA, reporting must also support compliance with Department of Migrant Workers documentation requirements, ensuring that the platform's records can serve as evidence of adherence to the regulatory framework.

Centralized data access is the primary advantage of cloud-based deployment, enabling authorized personnel to access recruitment information from any location with internet connectivity. Mkhize et al. (2025), in a systematic review of ninety studies, found that cloud-based systems offer significant advantages for cross-border organizations that require centralized access from multiple locations. Ahmad et al. (2025) examined cloud adoption determinants and found that accessibility and scalability are primary drivers. For FIRA, which operates across the Philippines and Morocco, cloud-based deployment through Supabase-hosted PostgreSQL and Supabase Storage ensures that applicant records, job information, and application statuses are accessible to authorized personnel regardless of their physical location. This eliminates the information fragmentation that characterizes local-storage-based practices, where files are accessible primarily through the personnel responsible for managing them.

Cloud-based data management extends centralized access to the storage, backup, security, and maintenance of recruitment data. Moreover, Foronda (2023), in a systematic review of Data Privacy Act implementation in Philippine institutions, found that cloud-based data management improves compliance with data protection requirements through centralized access control, encryption, and audit logging. Mkhize et al. (2025) found that cloud services reduce the operational burden of data

management by shifting infrastructure maintenance to the service provider. The proposed FIRA platform uses Supabase PostgreSQL as its primary database, with Prisma ORM facilitating interaction between the application and the database. Supabase Storage supports the storage of recruitment documents and related user files. This cloud-based architecture ensures that data management practices, including backup, encryption, and access logging, are handled through the cloud provider's infrastructure, which is subject to established security and compliance standards.

Cloud-based deployment offers benefits including scalability, accessibility, reduced infrastructure cost, and managed security, but it also introduces considerations related to data sovereignty, internet dependency, and vendor lock-in. Mkhize et al. (2025) found that organizations must weigh these factors against their operational requirements and regulatory obligations. Foronda (2023) emphasized that Philippine institutions subject to the Data Privacy Act must ensure that cloud providers comply with the Act's requirements for consent, purpose limitation, data minimization, and security. Ahmad et al. (2025) note that cloud adoption requires careful consideration of data classification, access control, and service-level agreements. Therefore, the proposed FIRA platform adopts a cloud-based approach because the agency's cross-border operations require centralized access, and the regulatory environment, including the Data Privacy Act of 2012 and the Department of Migrant Workers Rules, can be satisfied through appropriate access control, encryption, and audit logging implemented within the cloud infrastructure.

On-premise deployment, in contrast, hosts the recruitment platform on infrastructure that the agency owns, controls, and maintains within its own physical premises or

leased servers. This model offers direct hardware control, predictable data residency, and the absence of recurring cloud subscription costs, which can appeal to organizations with strict data sovereignty requirements or limited internet reliability. However, on-premise deployment requires significant upfront capital investment in servers, storage, and networking equipment, dedicated information technology personnel for maintenance, security patching, and backup operations, and it constrains multi-location access because remote users depend on virtual private network connections to a single physical site (Mkhize et al., 2025; Ahmad et al., 2025). For agencies operating across multiple jurisdictions, on-premise deployment also complicates disaster recovery and business continuity because redundancy must be provisioned and maintained by the agency itself. The proposed FIRA platform adopts a cloud-based approach because the agency's cross-border operations require centralized access from both the Philippine local agency and the Moroccan principal office, and the regulatory environment can be satisfied through appropriate access control, encryption, and audit logging implemented within the cloud infrastructure rather than through physical server control.

Responsive web design is the technical approach that allows a single web application to adapt its layout and functionality to different screen sizes and device contexts. Parpiev (2025) conducted an experimental comparison and found that responsive design significantly improves user satisfaction and task completion rates. Magdalenić and Luić (2025) applied design thinking principles to recruitment interfaces and argued that responsive design is essential for platforms serving multiple user roles with different device preferences. Washington (2023) examined responsive web design

in the context of recruitment platforms and found that mobile-first design is essential for applicant-facing interfaces. For the proposed FIRA platform, responsive design is implemented through Tailwind CSS, which provides a utility-first approach to building responsive interfaces that adapt to mobile, tablet, and desktop contexts. The platform also supports light and dark themes, English and Filipino language support, and adjustable text and display settings, ensuring that the interface is accessible to users with diverse needs and preferences.

Authentication and access control are the mechanisms that verify user identity and restrict access to information and functions appropriate to the user's assigned role. The Identity Management Institute (2024) describes role-based access control as the assignment of permissions to roles rather than individuals, ensuring that users can only access information and perform actions appropriate to their role. Foronda (2023) found that consent administration and records management are the most-failed compliance areas in Philippine institutions subject to the Data Privacy Act, underscoring the importance of robust access control. Sony et al. (2025) examined AI governance in human resource management and argued that access control must be designed into the system architecture. The National Privacy Commission (n.d.) establishes that the Data Privacy Act mandates access limitation to authorized personnel with legitimate purpose.

Secure recruitment records require encryption at rest and in transit, audit logging of access and modifications, and data subject rights implementation that allows applicants to access, correct, and request deletion of their personal data. Foronda (2023) emphasized that consent administration and records management are the most-failed

compliance areas in Philippine institutions subject to the Data Privacy Act. The proposed FIRA platform implements a three-role architecture, namely Applicant, Local Agency, and Employer, with discrete permission sets aligned with the Data Privacy Act's access-control mandate, ensuring that each role can access only the information and functions appropriate to its responsibilities. Audit logging maintains a record of who accessed what information and when, supporting compliance audits and investigations of unauthorized access.

Web accessibility ensures that the platform can be used by individuals with diverse abilities, including those using assistive technologies. Washington (2023) examined web accessibility in the context of recruitment platforms and found that accessibility compliance improves usability for all users, not only those with disabilities. The web.dev (n.d.) guidelines on responsive design and accessibility, including semantic HTML, ARIA labels, and keyboard navigation, inform the platform's implementation. The platform's accessibility features are evaluated as part of the usability evaluation described in the Evaluation theme of this chapter, ensuring that accessibility is treated as a quality characteristic rather than an afterthought.

Traditional recruitment methods, including manual recruitment, Facebook-based recruitment, and referral-based recruitment, remain prevalent in the Philippine OFW recruitment context despite the documented advantages of web-based approaches. Cal International Philippines (n.d.) documents that manual recruitment, characterized by paper-based application forms, in-person submission of documents, and physical filing of applicant records, persists in many Philippine agencies. Facebook-based recruitment uses social media posts and Facebook groups to advertise vacancies and capture

applications, often without structured data capture or status tracking. Referral-based recruitment relies on personal networks and word-of-mouth to source candidates, providing trust but limited reach. The transition from traditional to digital recruitment is documented by Mobo (2025), who examined AI adoption in Philippine human resource management and found that the transition is uneven, with larger organizations moving faster than smaller agencies. Hybrid recruitment methods, including recruitment caravans and job fairs, combine in-person outreach with digital capture, providing a bridge between traditional and digital approaches.

For FIRA, the use of Facebook for job postings, online forms for data capture, and spreadsheets for tracking reflects a hybrid approach that has not yet been consolidated into a single platform. The proposed FIRA platform addresses this fragmentation by providing a single environment in which job postings, application capture, screening, endorsement, and tracking are integrated, while preserving the agency's ability to conduct in-person outreach through caravans and job fairs as feeder channels whose outputs must be entered into the platform for processing. This consolidation eliminates the information fragmentation that characterizes the current hybrid approach, where information about the same applicant may be distributed across Facebook messages, email attachments, spreadsheets, and paper files.

Fil International Recruitment Agency (FIRA) is a Morocco-based, land-based recruitment agency that facilitates the overseas employment of Filipino workers. The agency operates across two jurisdictions, coordinating its local agency in the Philippines with its international principal office in Morocco. The scale of overseas Filipino employment provides the broader context for FIRA's operations: the Philippine Statistics

Authority (2025) estimates 2.19 million Overseas Filipino Workers in 2024, with cash remittances reaching approximately 262 billion pesos (Talabong, 2025). Morocco marked fifty years of diplomatic relations with the Philippines in 2025 (Philippine News Agency, 2025), underscoring the enduring framework within which FIRA operates. FIRA's recruitment activities are governed by the Department of Migrant Workers, which consolidated the Philippine Overseas Employment Administration, the Overseas Workers Welfare Administration, and other agencies under Republic Act No. 11641 (Department of Migrant Workers, n.d.).

Overseas employment of Filipino workers is a structural feature of the Philippine economy, with the Philippine Statistics Authority (2025) estimating 2.19 million Overseas Filipino Workers in 2024 and cash remittances reaching approximately 262 billion pesos (Talabong, 2025). The World Bank (2024) reports that remittance flows and labor mobility trends continue to underscore the importance of efficient, regulated recruitment channels for overseas Filipino workers. The Department of Migrant Workers (n.d.) regulates the overseas employment of Filipino workers through the 2023 DMW Rules, which consolidate the prior POEA rules governing agency licensing, accreditation, job-order approval, and deployment documentation. The overseas employment framework establishes that no Filipino worker shall be deployed overseas except under a verified job order, an approved employment contract, and through a licensed recruitment agency, as mandated by Republic Act No. 8042. For FIRA, this framework dictates the documentation and verification requirements that the proposed platform must support.

International employment opportunities for Filipino workers span a wide range of sectors and destination countries, with demand driven by labor shortages in host countries and the comparative skills and cost advantages of Filipino workers. The Arab News (2023) reports on labor cooperation deals between the Philippines and Morocco, illustrating the bilateral framework within which FIRA operates. Cal International Philippines (n.d.) documents that international employment opportunities vary by destination country and position, with additional requirements for specific sectors such as construction, hospitality, healthcare, and domestic work. The Philippine News Agency (2025) reports that Morocco marked fifty years of diplomatic relations with the Philippines in 2025, underscoring the enduring framework within which FIRA sources Filipino workers for Moroccan employers. The Department of Migrant Workers (2023) regulates the accreditation of foreign employers to ensure that they meet Philippine standards for worker protection, and the proposed FIRA platform supports this accreditation process through its Employer Module, where employers submit inquiries and provide job requirements for verification by FIRA personnel.

Recruitment agencies play a central role in the overseas employment of Filipino workers, serving as intermediaries between foreign employers and Filipino applicants. Respicio and Co. (n.d.) document that licensed agencies are responsible for sourcing, screening, endorsing, and deploying Filipino workers in compliance with the regulatory framework established by the Department of Migrant Workers. Cal International Philippines (n.d.) documents that the agency's role includes verifying the legitimacy of foreign employers, ensuring that job orders meet Philippine standards for worker welfare, screening applicants for qualification and fitness, endorsing qualified applicants

to foreign employers, and managing the deployment documentation required for overseas employment. The Department of Migrant Workers (2023) establishes the licensing framework that governs agency operations, including minimum paid-up capital, escrow deposits, surety bonds, and performance bonds. For FIRA, the recruitment agency role is implemented through the Local Agency Module of the proposed platform, which provides agency staff with the tools to manage job postings, screen applicants, endorse qualified candidates, and track deployment documentation.

The international recruitment process follows a sequence of stages from job order and vacancy management through application monitoring and deployment, with each stage generating structured data that feeds the next. Bikou et al. (2024) model the process as a pipeline in which job posting, application capture, screening, shortlisting, endorsement, and employer selection are interconnected stages rather than discrete events. Cal International Philippines (n.d.) and Universal Staffing Services (n.d.) document that the international recruitment process for Philippine land-based OFW agencies proceeds from job order and vacancy management through applicant registration, applicant screening and verification, applicant endorsement, employer interview and selection, and application monitoring and deployment. The proposed FIRA platform maps this process onto its five-status workflow, where each application progresses through New Application, FIRA Review, FIRA Approved, Employer Review, and Accepted or Declined. The following subtopics describe each stage of the international recruitment process.

Job order and vacancy management is the first stage of the international recruitment process, where foreign employers submit job requirements to the agency

and the agency verifies and posts the resulting vacancies. The Department of Migrant Workers (2023) regulates job-order approval to ensure that job orders meet Philippine standards for worker welfare, including wages, working conditions, and contract terms. Cal International Philippines (n.d.) documents that principal accreditation is a prerequisite for any job order, requiring verified corporate documentation, legal compliance, and job-order approval from the Department of Migrant Workers. The proposed FIRA platform manages job order and vacancy management through the Employer Module, where employers submit inquiries and provide job requirements, after which FIRA personnel manage their accounts and verify the submitted requirements before job postings are published. The platform's job posting function captures the job title, description, required skills, qualification requirements, and other metadata that feed the SBERT-based matching component. This structured capture ensures that the textual content of the job description is available for semantic comparison with applicant profiles, supporting the matching pipeline from the earliest stage of the recruitment process.

Applicant registration and screening are the stages where applicants submit their profiles and documents, and agency staff verify and screen them for qualification and fitness. Cal International Philippines (n.d.) documents that applicant registration typically involves the submission of personal data, educational background, employment history, skills and certifications, and qualification documents. The proposed FIRA platform manages applicant registration through the Applicant Module, where applicants submit their documents and qualification records. Applicant screening and verification are then performed by agency staff, who verify the authenticity and relevance of the submitted

documents before endorsing qualified applicants to the foreign employer. The textual content of the applicant's resume, profile, and qualification documents also serves as the input to the SBERT-based matching component, which produces a similarity score and ranked candidate list as decision-support information for the screening process.

Applicant endorsement and employer interview are the stages where verified applicants are presented to the foreign employer for final selection. Cal International Philippines (n.d.) documents that endorsement requires the transmission of complete applicant information, including qualifications, documents, and screening results, to the employer for review. The proposed FIRA platform manages this hand-off through the Employer Review status, where endorsed candidates' profiles are presented to the employer with the results of the matching component as decision-support information. Greenhouse (2025) reports that structured endorsement, where candidates are presented with consistent, comparable information, improves employer decision quality. The employer interview and selection stage follows, where the foreign employer conducts interviews with endorsed candidates and makes the final selection decision. The platform's matching results, including percentage similarity scores and ranked candidate lists, are presented as advisory information that supports the employer's review without replacing human judgment about factors not captured in textual profiles.

Application monitoring and deployment are the final stages of the international recruitment process, where the application status is tracked through acceptance or decline, and successful candidates are deployed to the foreign employer. The proposed FIRA platform tracks application status through the Accepted or Declined status, where the employer's selection decision is recorded and communicated to the applicant.

Deployment documentation is then tracked through the platform's application monitoring function, which maintains a record of the documents submitted, verified, and issued at each stage of the recruitment process. Del Rosario Law (2026) summarizes the DMW documentary framework, which includes the employment contract, medical certificate, passport, work visa, and Overseas Employment Certificate. The Department of Foreign Affairs (n.d.) handles passport and document authentication, while the Department of Justice (n.d.) manages departure formalities. The platform ensures that its records can serve as evidence of compliance with the documentary requirements for overseas deployment, supporting the agency's compliance with the regulatory framework established by the Department of Migrant Workers.

Stakeholders in international recruitment include the recruitment agency, the local agency, the international agency, the employer or principal, and the applicant or Overseas Filipino Worker. For FIRA, the recruitment agency is the licensed entity that bears regulatory responsibility for the recruitment process. The local agency, based in the Philippines, handles applicant sourcing, screening, and deployment documentation. The international agency, based in Morocco, handles employer relations and final selection coordination. The employer or principal is the foreign entity that submits job requirements and makes the final selection decision. The applicant or Overseas Filipino Worker is the individual who registers with the agency, submits documents, undergoes screening, and is deployed to the foreign employer. Each stakeholder has distinct needs and responsibilities that the proposed FIRA platform addresses through its role-based architecture, with discrete permission sets for each role aligned with the Data Privacy Act's access-control mandate.

FIRA operates across two jurisdictions, coordinating its local agency in the Philippines with its international principal office in Morocco. Locally, the agency sources, screens, and endorses Filipino applicants to foreign employers. Cal International Philippines (n.d.) documents that this two-jurisdiction structure is typical of Philippine land-based recruitment agencies, where the local agency handles applicant sourcing, screening, and deployment documentation, while the foreign principal handles employer relations and final selection. Lenasri and Shekhar (2025), in a systematic review of recruitment technologies, note that this cross-border structure introduces coordination challenges that single-jurisdiction platforms do not face. The World Bank (2024) reports that remittance flows and labor mobility trends continue to underscore the importance of efficient, regulated recruitment channels for overseas Filipino workers. For FIRA specifically, the separation of local and international operations creates an information asymmetry that the proposed platform addresses through centralized, role-based access for personnel in both jurisdictions.

At present, FIRA manages its recruitment activities through a combination of Facebook posts for job advertising, online forms for application capture, spreadsheets for applicant tracking, email for coordination with foreign employers, and local files for documentation. This hybrid approach, while functional, fragments applicant information across tools that are not simultaneously accessible to all authorized personnel. The FIRA recruitment workflow involves three primary categories of stakeholders: applicants, who register and submit profiles; local agency staff, who verify, screen, and endorse applicants; and foreign employers, who review endorsed candidates and make selection decisions. The workflow proceeds from job posting through application

capture, screening, endorsement, employer review, and final selection, with application status tracked through five stages: New Application, FIRA Review, FIRA Approved, Employer Review, and Accepted or Declined. Universal Staffing Services (n.d.) and Cal International Philippines (n.d.) document that this workflow is typical of Philippine land-based OFW recruitment agencies, where the absence of a centralized platform creates the operational and compliance challenges that the proposed FIRA platform seeks to address.

FIRA's existing recruitment practices reflect a hybrid of traditional and digital methods. The use of Facebook for job postings reflects the popularity of social media as a recruitment channel in the Philippines, where InsiderPH (2025) reports that 93 percent of Philippine companies hired talent through online platforms in 2024. However, the use of Facebook for recruitment also exposes applicants to scam risks, as documented by the Department of Migrant Workers (2026) and Cabato and Magsambol (2025). The proposed FIRA platform addresses this risk by providing a verifiable, regulated alternative to social-media-based recruitment, where the agency's legitimacy is established through its DMW license and the platform's records can serve as evidence of compliance with the regulatory framework.

The existing policies and involved agencies governing Philippine overseas recruitment constitute a regulatory framework of interlocking legal instruments and the agencies that administer them. The principal legal instruments are Republic Act No. 8042 (the Migrant Workers and Overseas Filipinos Act of 1995), Republic Act No. 10022 (its 2010 amendment), Republic Act No. 10173 (the Data Privacy Act of 2012), Republic Act No. 11641 (the Department of Migrant Workers Act of 2021), and the 2023 DMW Rules with

their 2024 implementation guidelines. The principal agencies are the Philippine Overseas Employment Administration (POEA), now consolidated under the Department of Migrant Workers (DMW); the Department of Foreign Affairs (DFA); the Overseas Workers Welfare Administration (OWWA); the Technical Education and Skills Development Authority (TESDA); and the statutory social protection bodies comprising PhilHealth, the Pag-IBIG Fund, and the Social Security System (SSS). These policies and agencies jointly define the documentation, screening, endorsement, and deployment requirements that the proposed FIRA platform must support through its application tracking and role-based access components.

The Philippine Overseas Employment Administration, now consolidated under the Department of Migrant Workers, was historically the primary regulatory body governing overseas employment. The Department of Migrant Workers (n.d.) was established by Republic Act No. 11641 to consolidate the functions of the POEA, OWWA, and other agencies under a single department. The Department of Migrant Workers (2023) issued the 2023 DMW Rules, which consolidate and update the prior POEA rules governing agency licensing, accreditation, job-order approval, and deployment documentation. Memorandum Circular No. 04, series 2024, provides implementation guidelines for private recruitment agencies, including foreign recruitment agencies.

The Department of Foreign Affairs (n.d.) handles passport issuance and document authentication through its consular offices and foreign posts. The Overseas Workers Welfare Administration (n.d.) provides welfare assistance programs and maintains OFW membership records. The Technical Education and Skills Development Authority (TESDA) (n.d.) provides skills certification through National Certificates that verify the

applicant's competence in specific trades. TESDA National Certificates are among the qualification documents that applicants submit and that the platform's applicant information management function must capture, as they are inputs to the screening and matching process.

Statutory social protection coverage for OFWs includes PhilHealth, Pag-IBIG Fund, and Social Security System contributions. PhilHealth (2024) established a 5 percent premium rate effective January 2024, up from the previous 4 percent. The Social Security System raised the contribution rate by one percentage point to 15 percent effective January 2025 (KPMG, 2025; PwC, 2026). The Pag-IBIG Fund (n.d.) maintains coverage requirements with monthly contributions ranging from 100 to 2,950 pesos, with 24 months of contributions required for housing loan eligibility. For FIRA, these statutory coverage requirements are components of the deployment documentation that the platform's records must track, as they are prerequisites for overseas deployment and worker welfare protection.

Applicant requirements for overseas deployment include a passport with at least six months validity, a work visa for the destination country, an authenticated employment contract, NBI clearance, TESDA National Certificates for skilled trades, medical examination results, and educational and experience credentials. The Department of Migrant Workers (2023) establishes the documentation framework for securing the Overseas Employment Certificate, which is the primary authorization document for overseas deployment. Cal International Philippines (n.d.) documents that applicant requirements vary by destination country and position, with additional requirements for specific sectors and roles. For FIRA, applicant requirements are captured through the

Applicant Module, where applicants submit their documents and qualification records, which are then verified by agency staff during the screening stage. The textual content of these documents, including resumes, job descriptions, skills, and qualification requirements, also serves as the input to the SBERT-based matching component.

Employer requirements include principal accreditation, job-order approval, verified job orders, and corporate and legal documentation demonstrating the employer's legitimacy and capacity to employ Filipino workers. Cal International Philippines (n.d.) documents that principal accreditation is a prerequisite for any deployment, requiring verified corporate documentation, legal compliance, and job-order approval from the Department of Migrant Workers. The Department of Migrant Workers (2023) regulates employer accreditation to ensure that foreign employers meet Philippine standards for worker protection. For FIRA, employer requirements are managed through the Employer Module, where employers submit inquiries and provide job requirements, after which FIRA personnel manage their accounts and verify the submitted requirements before job postings are published.

Recruitment agency requirements include minimum paid-up capital of 5 million pesos, an escrow deposit of 1.5 million pesos, a surety bond of 100,000 pesos, and a performance bond for the duration of the license. Respicio and Co. (n.d.) document these financial requirements, which are established by the Department of Migrant Workers (2023) to ensure that agencies have the financial capacity to fulfill their obligations to applicants and workers. The licensing framework also requires compliance with documentation, verification, and reporting standards. For FIRA, agency requirements are satisfied through compliance with the DMW regulatory framework, and

the proposed platform supports compliance by maintaining auditable records of recruitment activities, applicant verification, and deployment documentation.

Qualification documents include TESDA National Certificates, NBI clearance, educational credentials, experience certificates, and other documents that demonstrate the applicant's qualifications for a specific position. The Technical Education and Skills Development Authority (n.d.) provides skills certification through National Certificates that verify the applicant's competence in specific trades. These documents are the textual inputs consumed by the SBERT matching component, as they contain the information about skills, experience, and qualifications that the system compares with job requirements. Cal International Philippines (n.d.) documents that qualification documents must be verified for authenticity and relevance to the position before endorsement. The proposed FIRA platform captures qualification documents through the Applicant Module and makes them available to agency staff for verification and to the matching component for semantic comparison.

Documentary requirements for overseas deployment include the passport, work visa, and Overseas Employment Certificate trilogy, along with departure formalities managed by the Department of Justice. Del Rosario Law (2026) summarizes the DMW documentary framework, which includes the employment contract, medical certificate, and other documents specific to the destination country. The Department of Foreign Affairs (n.d.) handles passport and document authentication. The Department of Justice (n.d.) manages departure formalities. For FIRA, deployment documentation is tracked through the platform's application monitoring function, which maintains a record of the documents submitted, verified, and issued at each stage of the recruitment process.

This ensures that the platform's records can serve as evidence of compliance with the documentary requirements for overseas deployment.

Role-based access control assigns discrete permission sets to each user role, ensuring that users can only access information and perform actions appropriate to their assigned role. The Identity Management Institute (2024) describes role-based access control as essential for data security and regulatory compliance. Foronda (2023) found that consent administration and records management are the most-failed compliance areas in Philippine institutions subject to the Data Privacy Act, underscoring the importance of robust access control. Sony et al. (2025) examined AI governance in human resource management and argued that access control must be designed into the system architecture. The National Privacy Commission (n.d.) establishes that the Data Privacy Act mandates access limitation to authorized personnel with legitimate purpose. The proposed FIRA platform implements a three-role architecture, namely Applicant, Local Agency, and Employer, with discrete permission sets aligned with the Data Privacy Act's access-control mandate.

Republic Act No. 8042, the Migrant Workers and Overseas Filipinos Act of 1995, is the foundational statute governing the overseas employment of Filipino workers. The Act establishes the state policy that no Filipino worker shall be deployed overseas except under a verified job order, an approved employment contract, and through a licensed recruitment agency. The Act mandates bonds and escrow deposits for recruitment agencies, and criminalizes illegal recruitment, with penalties ranging from six years and one day to twelve years of imprisonment. Syndicated or large-scale illegal recruitment is classified as economic sabotage, reflecting the severity with which the

state treats the exploitation of Filipino workers. The Act establishes the framework for worker protection that the proposed FIRA platform supports through its verifiable, regulated recruitment environment, which distinguishes legitimate recruitment from the illegal practices that the Act seeks to eliminate. As a foundational statute, Republic Act No. 8042 predates the 2023 to 2026 publication window but is cited as the immutable legal basis for the regulatory framework within which the platform operates.

Republic Act No. 10022, enacted in 2010, amended Republic Act No. 8042 to strengthen the verification of overseas employment contracts. The amendment mandates that every employment contract be verified by the Philippine foreign post in the destination country, ensuring that the terms of employment meet Philippine standards for worker welfare. The amendment also elevated syndicated or large-scale illegal recruitment to the status of economic sabotage, providing a statutory basis for the severe penalties applied to large-scale illegal recruitment operations. The amendment reinforces the statutory mandate for verified job orders and contracts that the proposed FIRA platform supports through its documentation and verification functions. As a foundational amendment, Republic Act No. 10022 is cited as the legal basis for the contract verification requirements that govern the recruitment workflow.

Republic Act No. 10173, the Data Privacy Act of 2012, is the primary Philippine data protection law, governing the processing of personal data in the Philippines. The Act created the National Privacy Commission, established data-subject rights, and mandated the appointment of data protection officers, data processing system registration, privacy impact assessments, and security measures. The National Privacy Commission (n.d.) operationalizes the Act through the Data Privacy Act portal and the

2024 AI Advisory, which provides guidance on the use of artificial intelligence in processing personal data. Teodoro (2025), in a critical analysis of the Data Privacy Act, grounds the framework in the principles of legitimate purpose, proportionality, and transparency. Foronda (2023) found that consent administration and records management are the most-failed compliance areas. For FIRA, the Data Privacy Act dictates the role-based access control, encryption, audit logging, and data subject rights that the proposed platform implements. As a foundational statute, the Data Privacy Act is cited as the immutable legal basis for the data protection framework within which the platform operates.

The Department of Migrant Workers translates the statutory framework into operational rules through department orders and memoranda circulars. The 2023 DMW Rules consolidate and update the prior POEA rules governing agency licensing, accreditation, job-order approval, and deployment documentation. Memorandum Circular No. 04, series 2024, provides implementation guidelines for private recruitment agencies, including foreign recruitment agencies. The regulatory perimeter governing online recruitment is established by the 2023 DMW Rules and the 2024 implementation guidelines, which regulate agency conduct in online channels rather than prescribing an e-recruitment-specific standard. This regulatory gap, the absence of a dedicated DMW issuance on online recruitment platforms, is treated as a contribution of the proposed FIRA platform, which demonstrates how a regulated, verifiable online recruitment environment can be implemented within the existing DMW framework. This treatment is consistent with the academically honest approach of acknowledging the limits of the

literature search rather than over-claiming the existence of a specific issuance that could not be located.

RECRUITMENT APPLICATIONS, TOOLS, AND SYSTEM FEATURES

Recruitment applications encompass a range of digital platforms and tools that support the recruitment process, from job boards that connect applicants with employers to enterprise applicant tracking systems that manage the full hiring pipeline. Novaković et al. (2024) provide a three-class taxonomy of recruitment applications: job boards that aggregate and display job postings, enterprise applicant tracking systems that manage the end-to-end hiring workflow, and AI-augmented matching platforms that use machine learning to improve candidate-job alignment. Madanchian et al. (2024) note that the boundaries between these classes are increasingly blurred, as job boards add tracking features, applicant tracking systems incorporate AI matching, and dedicated matching platforms expand into workflow management. Persistence Market Research (n.d.) estimates the recruitment software market at \$3.20 billion in 2026, growing to \$5.56 billion by 2033 at an 8.2 percent compound annual growth rate, underscoring the commercial significance of this technology category.

Spoladore et al. (2026) argue that hybrid systems combining matching with structured workflow management consistently outperform single-signal approaches, suggesting that the most effective recruitment applications integrate rather than separate these functions. Lenasri and Shekhar (2025), in a systematic review identifying LinkedIn as the most dominant platform across sixty-two articles, note that the effectiveness of recruitment applications depends on integration with structured

screening workflows rather than isolated job advertising. Mobo (2025) examined AI adoption in Philippine human resource management and found growing interest in AI-assisted recruitment, though implementation remains limited to larger organizations. For FIRA, the overview of recruitment applications establishes the landscape against which the proposed platform's positioning, namely a regulated, agency-specific platform combining applicant tracking with semantic matching, can be evaluated.

Existing recruitment applications vary in their target users, feature sets, and suitability for the regulated agency-to-principal endorsement workflow that FIRA requires. The following platforms represent the dominant commercial options in the recruitment application market, each examined for its purpose, target users, major features, technologies, strengths, limitations, and relevance to the proposed platform.

JobStreet, operated by SEEK (SEEK Employer, n.d.), is the dominant job board in Southeast Asia, including the Philippines, with a focus on connecting jobseekers with employers through searchable job postings and applicant profiles. Jobma (2025) and Qureos (2026) describe JobStreet's recruitment process as employer-driven, where employers post vacancies and applicants submit applications through the platform. JobStreet (2024) provides recruitment-process guidance for both employers and applicants. InsiderPH (2025) reports that 93 percent of Philippine companies hired talent through online platforms in 2024, with JobStreet maintaining a leading position. JobStreet's strengths include its large applicant pool, brand recognition in the Philippine market, and structured job posting and application features. Its limitations for FIRA include its focus on direct employer-to-applicant matching rather than the agency-mediated endorsement workflow, the absence of DMW-regulated stages such

as job-order approval and deployment documentation, and the lack of semantic matching capabilities. JobStreet's keyword-based search represents the baseline against which FIRA's SBERT-based semantic matching is designed to improve.

LinkedIn, operated by Microsoft, is the dominant professional networking platform globally, with recruitment features including job postings, AI-assisted candidate sourcing, and employer branding. Lenasri and Shekhar (2025) identified LinkedIn as the most dominant platform across sixty-two articles in their systematic review. LinkedIn (2024) launched AI-assisted sourcing features that use machine learning to identify candidates matching employer requirements, and LinkedIn (2025) extended these features with additional AI capabilities. LinkedIn's strengths include its global professional network, AI-assisted sourcing, and rich candidate profiles. Its limitations for FIRA include the opacity of its matching logic, its focus on direct employer-to-applicant channels rather than the agency-mediated endorsement workflow, and its uneven penetration in OFW-relevant labor market segments. The platform's relevance to FIRA is as a reference point for AI-assisted sourcing, while its direct-to-employer model does not align with the agency-to-principal endorsement structure that the regulated OFW recruitment workflow requires.

Indeed is a job aggregation platform that collects job postings from multiple sources and provides search and application features for jobseekers. Indeed Hiring Lab (2024) provides labor-market intelligence based on platform data. Pin (2026b) reports a 4.7 percent applicant response rate across 635,000 tracked applications, a figure that underscores the challenge of achieving response in high-volume recruitment environments. Indeed (2026) maintained its aggregator model, though a 2024 policy

change ended organic traffic for already-indexed jobs, affecting how postings reach applicants. Indeed's strengths include its large job posting inventory and applicant reach. Its limitations for FIRA include its aggregator model, which does not provide the structured workflow management that the agency-to-principal endorsement process requires, and the low response rates that characterize high-volume aggregator platforms.

Workday is an enterprise human capital management platform that includes recruitment modules with applicant tracking, talent management, and AI-assisted features. Workday (2024) commissioned a Hanover Research survey on talent shortage and skills-based hiring, reflecting the platform's enterprise focus. Joveo (2025) describes Workday's acquisition of HiredScore, which added AI-assisted screening capabilities to the platform. Workday's strengths include its comprehensive human capital management suite, enterprise-grade features, and AI-assisted screening through HiredScore. Its limitations for FIRA include its enterprise-scale complexity and cost, which are prohibitive for smaller regulated agencies, and its focus on enterprise hiring rather than the specific agency-to-principal endorsement workflow. The platform's relevance to FIRA is as a reference point for enterprise-scale AI-assisted screening, while its scale and complexity make it unsuitable for the FIRA context.

Recruitment tools and applicant tracking systems focus specifically on the hiring workflow, providing features for pipeline management, candidate communication, and structured evaluation. The following platforms represent leading options in this category, examined for their relevance to the proposed FIRA platform.

Greenhouse is a structured hiring platform that emphasizes rubrics, scorecards, and interview kits to standardize the evaluation process. Greenhouse (2025) positions its structured hiring approach as a differentiator. People Managing People (2026) reviews Greenhouse's features, and Pin (2026a) reports a G2 Winter 2026 satisfaction score of 98 percent for Greenhouse. Greenhouse's strengths include its structured hiring approach, high user satisfaction, and interview kit features. Its limitations for FIRA include its focus on direct employer hiring rather than the agency-mediated endorsement workflow, and the absence of DMW-regulated stages. The platform's relevance to FIRA is as a reference point for structured evaluation, which the proposed platform incorporates through its role-based workflow and application status tracking. Additional context clarifies the platform's positioning. The purpose of Greenhouse is to standardize the hiring process through structured interview kits and decision rubrics that reduce interviewer bias and produce auditable hiring records. Its target users are in-house talent acquisition teams at mid-to-large enterprises that conduct high-volume professional hiring and require compliance documentation. The platform is built on a cloud-native web architecture with REST APIs for integration with human resource information systems, and it employs rule-based scorecard aggregation rather than machine-learning-based semantic matching, which distinguishes it from the algorithmic matching approach proposed for FIRA.

Lever is a combined applicant tracking system and candidate relationship management platform that emphasizes nurture pipelines for maintaining engagement with candidates over time. Lever (2026) documents its combined ATS and CRM features, and iSmartRecruit (2026) describes Lever's AI features, including fraud

prevention and interview transcript generation. Lever's strengths include its combined ATS and CRM functionality, nurture pipeline features, and AI-assisted capabilities. Its limitations for FIRA include its focus on direct employer hiring rather than the agency-mediated workflow, and the absence of DMW-regulated stages. The platform's relevance to FIRA is as a reference point for the nurture pipeline concept, which the proposed platform incorporates through its applicant pool management and waiting-list functionality for applicants not immediately matched to a vacancy. Additional context clarifies the platform's positioning. The purpose of Lever is to unify applicant tracking with candidate relationship management so that recruiters can maintain long-term engagement with passive candidates who are not yet ready to apply. Its target users are in-house recruiting teams at growth-stage technology companies that depend on proactive candidate sourcing rather than agency-mediated endorsement. The platform is built on a cloud-native architecture with a JavaScript-based single-page application frontend, REST and GraphQL APIs for third-party integration, and optional AI-assisted modules that perform interview transcript analysis and anomaly detection for fraud prevention rather than semantic candidate-job matching.

Workable is a recruitment platform targeting small and medium-sized enterprises, with AI-assisted shortlisting features that represent the closest commercial analogue to FIRA's SBERT-based matching. Mokahr (2026) reports that Workable's AI shortlisting achieves three times faster screening and 92 percent match to manual reviews. Workable's strengths include its AI-assisted shortlisting, SME-appropriate scale, and integration capabilities. Its limitations for FIRA include its focus on direct employer hiring rather than the agency-mediated workflow, and the absence of DMW-regulated stages.

The platform's relevance to FIRA is as the closest commercial analogue to the proposed SBERT-based matching component, providing a reference point for the AI-assisted shortlisting approach that the proposed platform adopts. Additional context clarifies the platform's positioning. The purpose of Workable is to reduce the screening burden on small and medium-sized enterprise recruiters through AI-assisted shortlisting that ranks applicants against the job description. Its target users are internal human resources generalists at small and medium-sized enterprises that lack dedicated recruiters and need an end-to-end hiring tool that scales from a few hires per year to several dozen. The platform is built on a cloud-native architecture with a React-based frontend, integrates with major job boards through standardized APIs, and employs proprietary machine-learning models that combine keyword overlap with limited semantic features, which represents the closest commercial analogue to the SBERT-based matching component proposed for FIRA, although it does not integrate with DMW-regulated deployment documentation.

A comparison of the reviewed recruitment applications and tools reveals differences across several dimensions relevant to the proposed FIRA platform. The comparison dimensions include matching technology, pipeline management, applicant tracking, deployment documentation, role-based access, and semantic matching transparency. Bevara et al. (2025) and Rosenberger et al. (2025) provide the semantic matching benchmarks against which the commercial platforms' keyword-based approaches are measured. Zhang et al. (2025) review explainable AI in talent recruitment and note that most commercial platforms treat matching results as opaque recommendations rather than transparent, auditable decision-support outputs. Nowak

and Pawłowska-Nowak (2026) demonstrate an interpretable multilayered approach with SHAP-based feature contributions. Figure 1 presents the comparison in tabular form.

Figure 1. *Comparison of Recruitment Applications and Tools on Dimensions Relevant to the Proposed FIRA Platform*

Platform	Matching Technology	Pipeline Mgmt	DMW Regulated Stages	Agency-Mediated	Transparency
JobStreet	Keyword-based	Limited	No	No	Opaque
LinkedIn	AI-assisted sourcing	Limited	No	No	Opaque
Indeed	Aggregator search	Minimal	No	No	Opaque
Workday	HiredScore AI screening	Enterprise ATS	No	No	Opaque
Greenhouse	Structured rubrics	Full ATS	No	No	Structured
Lever	AI-assisted nurture	ATS + CRM	No	No	Opaque
Workable	AI shortlisting	Full ATS	No	No	Opaque
Proposed FIRA	SBERT + Cosine + LightGBM	Five-status workflow	Yes	Yes	Auditable

The comparison reveals that no existing platform combines DMW-regulated deployment documentation, agency-to-principal endorsement workflow, role-based access for applicants, local agency staff, and foreign employers, and transparent semantic matching. Commercial platforms are built for mass-market or enterprise hiring, with keyword-based matching and direct employer-to-applicant channels. Research systems demonstrate semantic matching capabilities but in general or candidate-side

contexts rather than agency-side regulated cross-border contexts. This comparison establishes the feature gap that the proposed FIRA platform addresses through its integration of regulated workflow management with transparent semantic matching.

Benchmarking analysis across the reviewed platforms and the recruitment literature provides quantitative reference points for the proposed platform's expected performance. GoodTime (2026) provides time-to-hire benchmarks, and SHRM (2026) reports cost-per-hire averages of \$5,475 for non-executive positions. Pin (2026a, 2026b, 2026c) reports that AI-assisted recruitment tools can reduce time-to-hire by up to 82 percent, save approximately \$27,000 per hire, and improve hiring outcomes across multiple metrics. Tang et al. (2025) provide a systematic review of explainable person-job recommendation evaluation. Rafi et al. (2025) and Rojas et al. (2025) apply the ISO/IEC 25010 framework to recruitment system evaluation, with Rojas et al. (2025) conducting a bibliometric analysis of 1,613 publications on software quality.

Tracker-RMS (2026) reports that 86 percent of recruiters say applicant tracking systems reduced time-to-hire, while 94 percent reported a positive impact on hiring. Enad and Pabalan (2023) and Enad (2025) provide local Philippine context for recruitment system benchmarks. The benchmarking analysis establishes that the proposed FIRA platform's expected benefits, namely reduced time-to-hire, improved matching accuracy, and centralized recruitment management, are consistent with the quantitative evidence from existing platforms and the recruitment literature, while the platform's specific integration of regulated workflow with transparent semantic matching represents a contribution beyond the existing benchmark landscape.

The implications of the reviewed recruitment applications and tools for the proposed FIRA platform can be summarized across five dimensions. First, the matching technology gap: existing commercial platforms rely primarily on keyword-based matching, while research demonstrates that semantic matching using SBERT-family models offers significant improvements in retrieval accuracy (Ajjam et al., 2025; Kurek et al., 2024; Rosenberger et al., 2025). The proposed platform's SBERT-based matching addresses this gap. Second, the pipeline management gap: existing platforms treat the agency, the employer, and the applicant as separate accounts rather than participants in a single end-to-end recruitment pipeline (Novaković et al., 2024). The proposed platform's five-status workflow addresses this gap.

Third, the transparency gap: existing platforms treat matching results as opaque recommendations rather than transparent, auditable decision-support outputs (Zhang et al., 2025). The proposed platform's matched and missing skills identification addresses this gap. Fourth, the regulatory gap: existing platforms do not incorporate DMW-regulated deployment documentation, job-order approval, or the agency-to-principal endorsement workflow. The proposed platform's role-based architecture and documentation tracking address this gap. Fifth, the contextual gap: existing platforms are designed for mass-market or enterprise hiring rather than for smaller, regulated, cross-border agencies. The proposed platform's agency-specific design addresses this gap. Collectively, these implications establish that the proposed FIRA platform's contribution lies not in any single technological advance but in the integration of existing algorithmic components into a regulated, agency-specific recruitment environment.

CANDIDATE-JOB MATCHING

Candidate-job matching is the process of evaluating the alignment between an applicant's qualifications and a job's requirements to produce a ranking or recommendation that supports recruitment decisions. Ajjam et al. (2025) frame the matching problem as one that requires an algorithmic pipeline transforming heterogeneous applicant and job texts into a defensible ordering of candidates. Rosenberger et al. (2025) describe the matching model as a trained artifact that captures the semantic relationships between applicant qualifications and job requirements. Spoladore et al. (2026) argue that hybrid algorithmic systems combining semantic representation with structured comparison consistently outperform single-signal approaches. The proposed FIRA platform adopts a hybrid approach, integrating SBERT semantic embeddings, cosine similarity scoring, and LightGBM ranking within a single pipeline that produces transparent, auditable matching results as decision-support information for recruitment personnel and employers.

The methodology of candidate-job matching encompasses the alignment of applicant qualifications with job requirements through a structured process of text preprocessing, semantic representation, similarity computation, and ranking. Jirjees (2025) and Li (2024) describe features including experience years, skill-match ratio, and education alignment as inputs to the matching process. Sugiura (2024) and Zhao et al. (2024) describe user-item features, behavioral features, contextual features, and content features that contribute to matching accuracy. For the proposed FIRA platform, candidate qualification alignment involves comparing the textual content of an applicant's resume, profile, and qualification documents with the textual content of a job

description, skills list, and qualification requirements. Job requirement alignment involves the reverse perspective: ensuring that the job's requirements are fully represented in the matching process, with no requirement omitted or underweighted. Together, these alignment processes produce a feature vector that captures both the overall semantic alignment and the specific requirement satisfaction.

The importance of candidate-job matching lies in its capacity to reduce time-to-hire, improve the quality of hires, and provide a defensible basis for recruitment decisions. Koman et al. (2024) found that effective matching reduces recruitment cycle time by enabling agencies to focus their screening effort on the most promising candidates. Ajjam et al. (2025) demonstrated that semantic matching outperforms keyword matching on standard retrieval metrics, indicating that conventional matching leaves substantial improvement unrealized. Aricheta, Santiano, and Ayo (2026) frame the Philippine online job market as suffering from information overload that effective matching can address. The proposed FIRA platform's matching component is designed to produce ranked candidate lists with associated similarity scores and matched and missing skill identification, providing recruitment personnel and employers with decision-support information that supports rather than replaces human evaluation.

A review and comparison of matching approaches reveals a spectrum from keyword-based matching through statistical term weighting to semantic embedding approaches. Keyword-based matching, the simplest approach, treats vocabulary overlap as the primary signal of alignment, missing candidates who use different but semantically equivalent terms. Term frequency-inverse document frequency (TF-IDF) improves on raw keyword counting by weighting terms according to their discriminative

power, but remains a lexical approach. Word2Vec replaces term counting with learned dense vector representations, capturing some semantic relationships through the distributional hypothesis, but is limited by its static nature: each word receives a single vector regardless of context. SBERT and other transformer-based approaches produce contextual embeddings that vary with context, disambiguating polysemous terms and capturing the full semantic content of sentences and paragraphs. The proposed FIRA platform adopts SBERT as the semantic component of its matching pipeline, with keyword matching and TF-IDF serving as baselines against which the semantic approach is evaluated.

SBERT-based semantic matching is the approach the proposed FIRA platform adopts for generating semantic representations of applicant and job textual information. Reimers and Gurevych (2019) introduced SBERT as a modification of BERT that produces semantically meaningful sentence embeddings suitable for efficient comparison through cosine similarity. Unlike the cross-encoder approach, which requires running a full pairwise BERT comparison for every candidate-job pair, the Siamese approach encodes texts independently and then compares the resulting vectors, reducing the computational cost from quadratic to linear in the number of comparisons. Rosenberger et al. (2025) and Bevara et al. (2025) apply SBERT-based architectures to recruitment matching, confirming the suitability of the Siamese approach for candidate-job comparison. For the proposed FIRA platform, SBERT enables the efficient comparison of applicant and job embeddings that the matching pipeline requires.

Cosine similarity is the measure by which the proposed platform compares the SBERT-generated embeddings of applicant and job information, computing the directional alignment between vectors as a basis for ranking. Manning, Raghavan, and Schütze (2008) establish cosine similarity as a foundational measure in information retrieval, and Reimers and Gurevych (2019) adopt it as the standard comparison measure for SBERT-generated embeddings. Rieon et al. (2026) report that cosine similarity is the standard comparison measure for SBERT-based candidate matching. For the proposed FIRA platform, cosine similarity is applied after SBERT embedding generation, producing a similarity score for each applicant-job pair that serves as an input to the LightGBM ranking component.

LightGBM is the gradient-boosted decision-tree framework that the proposed platform adopts for ranking candidates based on integrated features. Ke et al. (2017) introduced LightGBM, describing techniques for efficient training including Gradient-based One-Side Sampling and Exclusive Feature Bundling. Li (2024) applied LightGBM to enterprise recruitment and reported strong performance with efficient training, with results of 78.6 percent accuracy, 75.4 percent precision, 80.1 percent recall, and 77.7 percent F1 score in recruitment matching. Unlike SBERT, LightGBM does not directly supply a semantic representation of text; its role is to combine already-constructed features, such as semantic similarity and structured requirement comparisons, within a supervised prediction or ranking task. The proposed FIRA platform adopts the ranking objective, where LightGBM produces scores used to order candidates within a job group rather than to estimate a probability of qualification.

The ranking and recommendation approach adopted by the proposed FIRA platform combines the semantic similarity score with structured features in a supervised learning-to-rank framework. Kannen et al. (2024) examined pointwise and pairwise learning-to-rank objectives, which determine how the model is trained to produce rankings. Shen et al. (2024) describe learning-to-retrieve as the production standard for large-scale matching, where embedding retrieval, similarity scoring, and learning-to-rank are combined. For the proposed FIRA platform, the LightGBM model is trained on applicant-job pairs with relevance labels and produces a ranking score for each new pair. The scores are used to order candidates within a job group, producing the ranked candidate list that is presented as decision-support information to recruitment personnel and employers. The ranking is advisory, and the final selection decision remains with authorized human users who can evaluate factors not captured in the feature vector.

Processing speed and efficiency are critical for a recruitment platform that must handle high volumes of applicants and job postings. Liu et al. (2025) report that batched, GPU-accelerated SBERT embedding can process approximately 1,000 resumes per minute. For normalized embeddings, cosine similarity equals the dot product, which is a single matrix multiplication operation that can be computed efficiently for all applicant-job pairs simultaneously. LightGBM, with its leaf-wise growth, histogram-based splitting, Gradient-based One-Side Sampling, and Exclusive Feature Bundling, achieves faster training and inference than XGBoost while maintaining comparable or better accuracy (Li, 2024). The embedding generation is independent for each applicant and job, meaning that stored embeddings can be reused when the

source text has not changed, providing computational efficiency for the comparison stage. Together, these characteristics enable the proposed platform to handle the volume of applicants that a Philippine land-based OFW recruitment agency typically processes.

The challenges and limitations of candidate-job matching include mismatched job placement, bias and fairness concerns, and the difficulty of explaining automated decisions. Mismatched job placement occurs when the matching algorithm produces a ranking that does not align with the actual suitability of candidates for the position, often due to incomplete or inaccurate data, or to the algorithm's inability to capture factors not represented in textual profiles. Aricheta, Santiano, and Ayo (2026) document the information overload and keyword-based search limitations that contribute to mismatched job placement in the Philippine online job market. The proposed FIRA platform addresses mismatched job placement through the integration of semantic and structured features, the matched and missing skills identification that makes the basis for each match traceable, and the decision-support framing that retains human authority over final decisions.

Bias and fairness considerations are central to the design of any automated matching system, particularly one used in recruitment where decisions affect individuals' livelihoods. Fabris et al. (2025) catalogued bias sources in algorithmic hiring, including measurement bias, representation bias, and aggregation bias, and identified fairness requirements that matching systems must address. Rigotti and Fosch-Villaronga (2024) emphasize the need for accountability and the right to challenge automated decisions in AI-based recruitment. The proposed FIRA platform addresses bias and fairness through

several mechanisms: historical acceptance decisions are not automatically treated as ground truth, as decisions may reflect unrecorded constraints or previous bias; protected attributes are not used as input features; the matched and missing skills identification provides a traceable basis for each match that can be audited for bias; and the decision-support framing retains human authority over final decisions, providing a human-in-the-loop safeguard against algorithmic bias.

Mitigation strategies for bias include debiasing and reweighting. Debiasing involves modifying the training data or the model to reduce the influence of biased patterns, while reweighting involves adjusting the weights of training examples to compensate for underrepresented groups. Fabris et al. (2025) describe these strategies in detail. For the proposed FIRA platform, the choice of mitigation strategy depends on the specific bias sources identified during the evaluation phase, and the platform's documentation requirements ensure that the chosen strategy is recorded as part of the matching method. The platform's transparency features, including feature importance analysis through approaches such as SHAP, support the identification of bias sources and the evaluation of mitigation effectiveness.

Explainability and transparency in matching results are essential for recruitment decisions that affect individuals' livelihoods. Zhang et al. (2025) review explainable AI in talent recruitment and note that most commercial platforms treat matching results as opaque recommendations rather than transparent, auditable decision-support outputs. Nowak and Pawłowska-Nowak (2026) demonstrate an interpretable multilayered approach with SHAP-based feature contributions. Jamal (2024) applied SHAP, LIME, and permutation feature importance to recruitment data and found that feature

importance analysis increases recruiter trust and reduces time-to-decision. For the proposed FIRA platform, transparency is achieved through the matched and missing skills identification, which provides a traceable basis for each match by showing which required skills the applicant possesses and which are absent, and through feature importance analysis that identifies which features drove the ranking. The platform's transparency features support the recruiter's ability to understand, trust, and challenge the matching results.

Human-in-the-loop and recruiter override mechanisms are the safeguards that retain human authority over final hiring decisions. Lazaros et al. (2026) taxonomize human-in-the-loop patterns in AI-based decision systems, identifying the design choices that determine where human judgment enters the pipeline. Rigotti and Fosch-Villaronga (2024) emphasize the need for accountability and the right to challenge automated decisions in AI-based recruitment. For the proposed FIRA platform, the human-in-the-loop safeguard is implemented through the decision-support framing: the matching component produces ranked candidate lists with associated similarity scores and matched and missing skill identification as advisory information, while the final selection decision remains with authorized human users. The platform's role-based access control ensures that only authorized personnel can make selection decisions, and the audit logging records who made each decision and when, supporting accountability and the right to challenge automated decisions.

ALGORITHM AND MODEL

An algorithm is a computational procedure that transforms inputs into outputs through a defined sequence of operations, while a model is a trained machine learning artifact that captures patterns from data to make predictions or rankings. Ajjam et al. (2025) frame the candidate-job matching problem as one that requires an algorithmic pipeline transforming heterogeneous applicant and job texts into a defensible ordering of candidates. Rosenberger et al. (2025) describe the matching model as a trained artifact that captures the semantic relationships between applicant qualifications and job requirements. Spoladore et al. (2026) argue that hybrid algorithmic systems combining semantic representation with structured comparison consistently outperform single-signal approaches.

The proposed FIRA platform adopts a three-component algorithmic pipeline: SBERT for generating semantic embeddings of applicant and job textual information, cosine similarity for computing the directional alignment between embeddings, and LightGBM for ranking candidates based on integrated features. This pipeline transforms the inputs, namely resumes, profiles, job descriptions, skills, and qualification requirements, into outputs, namely ranked candidate lists with associated similarity scores and matched and missing skill identification, that serve as decision-support information for recruitment personnel and employers. The pipeline is designed to be transparent, auditable, and subject to human override, consistent with the decision-support framing established in Chapter 1 and the ethical requirements identified in the reviewed literature.

The matching process encompasses the sequence of operations from input preparation through final ranking, beginning with traditional and keyword-based

matching approaches that provide the baseline against which the proposed SBERT-based semantic matching is evaluated. Manual resume screening, the simplest approach, involves human recruiters reviewing each resume against the job requirements, producing a subjective and unscalable ranking. Keyword-based matching automates this process by counting the overlap between keywords in the resume and the job description, treating vocabulary as the primary signal of alignment. Ajjam et al. (2025) found that keyword-based matching leaves substantial improvement unrealized compared to semantic approaches, as it cannot capture synonyms, related concepts, or the contextual meaning of skills and experience. The limitation of keyword matching is that it treats vocabulary as the primary signal of alignment, missing candidates who use different but semantically equivalent terms to describe relevant qualifications.

Term frequency-inverse document frequency improves on raw keyword counting by weighting terms according to their frequency in a document relative to their frequency across the document collection. The TF-IDF weight for term t in document d within a collection D is given by the product of the term frequency and the inverse document frequency, as shown in Figure 2. Sutriawan et al. (2025) compared TF-IDF with Word2Vec, FastText, BERT, and GPT across matching tasks and found that TF-IDF, while outperforming raw keyword matching, is consistently surpassed by contextual embedding approaches. The limitation of TF-IDF is that it remains a term-frequency-based approach, capturing lexical rather than semantic alignment. TF-IDF serves as an intermediate baseline between keyword matching and semantic embedding, establishing the performance improvement that semantic approaches offer over statistical term-weighting methods.

$$tfidf_{(t,d,D)} = tf_{(t,d)} \times idf_{(t,D)}$$

Figure 2. *TF-IDF Weight for Term t in Document d within Collection D*

Word2Vec replaces term counting with learned dense vector representations of words, capturing semantic relationships through the distributional hypothesis that words appearing in similar contexts have similar meanings. Mikolov et al. (2013) introduced Word2Vec, which learns embeddings through either a continuous bag-of-words or a skip-gram architecture. The skip-gram objective maximizes the log probability of context words given a target word, as shown in Figure 3. Abdelmotaleb, Sorkhabi, and Hassaan (2025) examined the impact of embedding choice on resume-job matching performance and found that Word2Vec, while improving over TF-IDF, is limited by its static nature: each word receives a single vector regardless of context, so polysemous terms cannot be disambiguated. Sutriawan et al. (2025) found that Word2Vec is surpassed by contextual embedding approaches that vary with context. The limitation of Word2Vec is that it captures word-level rather than sentence-level semantics, and its static embeddings cannot disambiguate polysemous terms.

$$L = \frac{-1}{T} \sum_{t=1}^T \sum_{\substack{j \neq 0 \\ -c \leq j \leq c}} \log p(w_{t+j} | w_t)$$

Figure 3. *Word2Vec Skip-Gram Objective for Context Window c and Vocabulary Size T*

The limitations of keyword matching, TF-IDF, and static embeddings such as Word2Vec can be summarized as follows: keyword matching treats vocabulary as the primary signal of alignment, missing semantically equivalent terms; TF-IDF improves on

keyword counting but remains a lexical approach; and Word2Vec captures word-level rather than sentence-level semantics and cannot disambiguate polysemous terms. These limitations motivate the adoption of contextual semantic embedding approaches, particularly SBERT, which produces sentence-level embeddings that vary with context and capture the full semantic content of phrases and sentences. The proposed FIRA platform adopts SBERT as the semantic component of its matching pipeline, with keyword matching, TF-IDF, and Word2Vec serving as baselines against which the semantic approach is evaluated.

The matching methods and algorithm system architecture of the proposed FIRA platform encompass the sequence of operations from input preparation through final ranking. Shen et al. (2024) describe learning-to-retrieve as the production standard for large-scale matching, where embedding retrieval, similarity scoring, and learning-to-rank are combined. The proposed FIRA platform adopts this architecture, with the matching component that applies the SBERT-plus-cosine-similarity-plus-LightGBM pipeline. Ajjam et al. (2025) developed an AI-driven semantic similarity-based matching framework that produces ranked candidate lists and similarity scores as decision-support information. The architecture is designed to be transparent, auditable, and subject to human override, with each stage producing intermediate outputs that can be inspected and verified.

The AI-powered resume and job description matcher is the integrated component that applies the matching pipeline to applicant and job textual information. Bevara et al. (2025), in developing Resume2Vec, observed that transforming applicant tracking systems with semantic representations requires rethinking how applicant data is

structured and compared, not merely digitizing existing forms. Rosenberger et al. (2025) and Rieon et al. (2026) demonstrate the application of SBERT-based matching to recruitment contexts. For the proposed FIRA platform, the matcher takes as input the textual content of an applicant's resume, profile, and qualification documents, and the textual content of a job description, skills list, and qualification requirements, and produces as output a ranked candidate list with associated similarity scores and matched and missing skill identification.

Text preprocessing and normalization prepare the raw textual input for the matching pipeline. The preprocessing steps include tokenization according to the SBERT model's tokenizer, lowercasing, removal of formatting artifacts, and normalization of skill and qualification terms. Liu et al. (2025) describe preprocessing pipelines for SBERT-based matching. Long documents are handled through section-based encoding or controlled chunking, with the chosen strategy evaluated using representative documents rather than assuming that truncation preserves all relevant information. The preprocessing approach is documented as part of the matching method, as it affects the model's suitability for the FIRA context.

Sentence-BERT is the primary semantic component of the proposed FIRA platform, responsible for generating the vector representations of applicant and job textual information that the matching pipeline compares. Reimers and Gurevych (2019) introduced SBERT as a modification of BERT that produces semantically meaningful sentence embeddings suitable for efficient comparison through cosine similarity. The SBERT architecture enables the efficient comparison of applicant and job embeddings

that the matching pipeline requires, reducing the computational cost from quadratic to linear in the number of comparisons.

BERT, or Bidirectional Encoder Representations from Transformers, is the foundation upon which SBERT is built. Devlin et al. (2018) introduced BERT as a multi-layer Transformer encoder that uses masked language modeling and next sentence prediction to learn bidirectional contextual representations of text. The Transformer architecture, characterized by self-attention mechanisms, allows every token to attend to every other token in both directions, capturing the contextual meaning of each word relative to the full input. Eltahier et al. (2025) describe BERT's architecture in the context of recruitment applications, noting that the multi-layer encoder and self-attention mechanism enable the model to capture complex semantic relationships. Sutriawan et al. (2025) benchmarked BERT against other embedding models and found that its bidirectional context modeling improves semantic alignment. Guder et al. (2026) examine the computational cost of self-attention and the trade-offs involved in applying BERT-based models to large-scale matching tasks.

Sentence-BERT adapts the BERT architecture using Siamese and triplet network structures, as introduced by Reimers and Gurevych (2019). The Siamese architecture consists of two or more BERT networks with shared weights that process different inputs independently, producing fixed-size sentence embeddings that can be compared efficiently. Unlike the cross-encoder approach, which requires running a full pairwise BERT comparison for every candidate-job pair, the Siamese approach encodes texts independently and then compares the resulting vectors, reducing the computational cost from quadratic to linear in the number of comparisons. The triplet network structure

extends this approach by using three inputs, namely an anchor, a positive example, and a negative example, to train the model to produce embeddings where similar texts are close together and dissimilar texts are far apart in the vector space. Rosenberger et al. (2025) and Bevara et al. (2025) apply SBERT-based architectures to recruitment matching, confirming the suitability of the Siamese approach for candidate-job comparison.

The Siamese network is trained using a contrastive objective function that pulls semantically similar sentence pairs closer together in the embedding space while pushing dissimilar pairs apart. Reimers and Gurevych (2019) train the Siamese architecture on in-domain sentence pairs using the softmax loss objective shown in Figure 4. The objective operates on the cosine similarity between the two sentence embeddings, scaled by a temperature parameter alpha, and applies the softmax function across a set of candidate labels. For each anchor sentence with the correct matching sentence and a batch of negative candidates, the objective maximizes the log-likelihood of the correct pair. Abdelmotaleb, Sorkhabi, and Hassaan (2025) confirm that this contrastive training objective produces embeddings whose cosine similarity correlates well with semantic alignment in recruitment matching tasks, supporting the use of cosine similarity as the downstream comparison measure.

$$L = -\log \frac{\exp(\alpha \cdot \cos(s_i, s_{j_*}))}{\sum_{j=1}^N \exp(\alpha \cdot \cos(s_i, s_j))}$$

Figure 4. *S-BERT Siamese Softmax Loss Objective for Anchor Sentence with Correct Match and N Candidates*

where s_i is the embedding of the anchor sentence (e.g., a job description), s_{j+} is the embedding of the correct matching sentence (e.g., the qualified applicant's resume), s_n ranges over a batch of N candidate sentences (including the correct match and $N-1$ negatives), α is the inverse-temperature scaling parameter, and $\cos(\cdot, \cdot)$ is the cosine similarity between the two embeddings. The objective rewards high cosine similarity between the anchor and the correct match while penalizing high similarity with negative candidates, training the network to produce a vector space in which cosine similarity reflects semantic alignment. This formulation is the standard training objective for the SBERT Siamese architecture and is the mathematical basis for the cosine similarity scoring that the proposed FIRA platform adopts in its matching pipeline.

Sentence embeddings are the fixed-dimensional numerical vectors that SBERT generates for each input text, capturing its semantic content in a form suitable for comparison. Reimers and Gurevych (2019) establish that SBERT produces embeddings where related texts receive similar vectors despite differences in wording, enabling the system to identify semantic alignment that keyword-based approaches miss. Rosenberger et al. (2025) demonstrate that domain-adapted SBERT embeddings improve matching accuracy over generic approaches. For the proposed FIRA platform, the sentence embedding dimension and maximum input length are determined by the selected SBERT model checkpoint. The embedding process involves tokenization, transformer encoding, and a pooling operation that produces the fixed-size vector. The embeddings are normalized to unit length to simplify cosine similarity computation, with the raw embeddings retained for audit and analysis purposes. Figure 5 presents the SBERT embedding generation process in pseudocode form.

Figure 5. Pseudocode for SBERT Sentence Embedding Generation

```
FUNCTION generate_embedding(input_text, sbert_model):  
    // Step 1: Preprocess and tokenize the input text  
    clean_text <- preprocess(input_text)  
    tokens <- sbert_model.tokenize(clean_text)  
    IF length(tokens) > sbert_model.max_input_length THEN  
        tokens <- truncate(tokens, sbert_model.max_input_length)  
    END IF  
    // Step 2: Encode tokens through the Transformer encoder  
    token_embeddings <- sbert_model.encode(tokens)  
    // Step 3: Apply pooling operation to produce a fixed-size vector  
    sentence_embedding <- sbert_model.pool(token_embeddings)  
    // Step 4: L2 normalize the embedding for cosine similarity  
    norm <- sqrt(sum(sentence_embedding * sentence_embedding))  
    IF norm > 0 THEN  
        sentence_embedding <- sentence_embedding / norm  
    END IF  
    RETURN sentence_embedding  
END FUNCTION
```

Transfer learning and pretrained models enable SBERT to leverage knowledge learned from large text corpora and adapt it to the recruitment domain. Eltahier et al. (2025) found that transfer learning with pretrained SBERT models achieves high F1 scores in recruitment matching tasks, with learning rate being a critical hyperparameter. Strangmann et al. (2024) examined knowledge transfer and fine-tuning for large language models, providing guidance on adapting pretrained models to specific domains. For the proposed FIRA platform, transfer learning involves selecting a pretrained SBERT model, fine-tuning it on recruitment-specific data if available, and evaluating its performance on held-out recruitment matching tasks. The pretrained model provides a starting point that captures general semantic relationships, while fine-tuning adapts it to the specific vocabulary, structure, and alignment patterns of recruitment texts. The choice of pretrained model, including its training data, supported

languages, embedding dimension, and input length, is documented as part of the matching method, as these properties affect the model's suitability for the FIRA context.

Multilingual and cross-lingual considerations are significant for the proposed FIRA platform, which processes English and Filipino text and may encounter job descriptions and resumes in multiple languages. Reimers and Gurevych (2020) describe knowledge distillation for multilingual sentence embeddings, showing that multilingual capability comes from a particular training approach and data rather than from the SBERT architecture alone. Michail and Clematide (2025) examined adversarial cross-lingual approaches and found that performance varies across language pairs. Aricheta, Santiano, and Ayo (2026) developed HanapBuh.AI for the Philippine online job market, providing local context for cross-lingual matching. The choice of multilingual model and its evaluation on Filipino text is documented as part of the matching method, as cross-lingual variance in performance affects the platform's effectiveness.

Previous applications of SBERT in recruitment and related matching contexts establish its fitness for the candidate-job matching task. Reimers and Gurevych (2019) originally validated SBERT on semantic textual similarity benchmarks, where it outperformed BERT's pooled output by a wide margin while reducing inference time by approximately ninety percent. Ajjam et al. (2025) applied SBERT within an AI-driven semantic similarity matching framework for recruitment and reported consistent improvements over keyword-based comparison on standard retrieval metrics. Kurek et al. (2024) evaluated SBERT-family models on a candidate retrieval task and found that the correct candidate appeared within the top-500 results far more reliably than under a keyword baseline. Rosenberger et al. (2025) applied domain-adapted SBERT embeddings to recruitment

matching and demonstrated accuracy gains over generic embeddings. These prior applications confirm that SBERT is a validated foundation for the semantic component of the proposed FIRA platform's matching pipeline.

A comparison of SBERT with other embedding models reveals its suitability for the candidate-job matching task. Sutriawan et al. (2025) compared TF-IDF, Word2Vec, FastText, BERT, and GPT across matching tasks and found that contextual embedding approaches, particularly SBERT, consistently outperform static and lexical approaches. Khan et al. (2026) benchmarked BERT, RoBERTa, DistilBERT, and SBERT for candidate-job matching. Jirjees (2025) describes BERT embeddings for job matching. Rieon et al. (2026) report that SBERT combined with cosine similarity achieves strong retrieval performance. The advantages of SBERT include its sentence-level semantic representation, its efficiency through the Siamese architecture, and its compatibility with cosine similarity. The limitations of SBERT include its dependence on the pretrained model's training data, its computational cost relative to simpler approaches, and its cross-lingual variance in performance. For the proposed FIRA platform, SBERT's advantages outweigh its limitations, and the limitations are addressed through model selection, fine-tuning, and the decision-support framing that retains human authority over final decisions.

Cosine similarity is the measure by which the proposed platform compares the SBERT-generated embeddings of applicant and job information, computing the directional alignment between vectors as a basis for ranking. Manning, Raghavan, and Schütze (2008) establish cosine similarity as a foundational measure in information retrieval, and Reimers and Gurevych (2019) adopt it as the standard comparison

measure for SBERT-generated embeddings. Cosine similarity measures the cosine of the angle between two nonzero vectors, providing a measure of their directional alignment that is independent of their magnitude. For text embeddings, it provides a numerical comparison between the representation of an applicant's information and that of a job description. Its role is distinct from SBERT: SBERT constructs the vectors, while cosine similarity compares them. The measure ranges from negative one to one for nonzero real-valued vectors, with a value of one indicating identical direction, zero indicating orthogonality, and negative one indicating opposite direction. For normalized embeddings, the dot product equals the cosine similarity, which simplifies computation. Zero vectors require explicit handling because the denominator would be zero.

Vector similarity measurement encompasses several approaches, including Euclidean distance, Jaccard similarity, dot product, and cosine similarity. Manning, Raghavan, and Schütze (2008) compare these measures in the context of information retrieval. Cosine similarity is preferred for text embeddings because it is scale-invariant, bounded, and efficient to compute for normalized vectors. Steck et al. (2024) examined the interpretation and meaningfulness of cosine similarity of embeddings, noting that while cosine similarity provides a numerical comparison, its absolute value is not inherently meaningful without context. Muennighoff et al. (2023) provide additional guidance on the properties of cosine similarity for embedding comparison.

Cosine similarity scoring computes the similarity between applicant and job embeddings as a basis for ranking. For two nonzero vectors a and b , cosine similarity is defined as the dot product of the vectors divided by the product of their magnitudes, as shown in Figure 6. Rieon et al. (2026) report that cosine similarity is the standard

comparison measure for SBERT-based candidate matching. Kurek et al. (2024) used cosine similarity with SBERT-family models and found that an SBERT-family model surfaced the correct candidate within the top-500 results far more reliably than a keyword baseline. Steck et al. (2024) caution that cosine similarity of embeddings can be affected by how representations are constructed, and that the absolute value of the score is not a probability of qualification.

$$\cos(\theta) = \frac{a \cdot b}{||a|| \times ||b||} = \frac{\sum_{i=1}^n a_i b_i}{\sqrt{\sum_{i=1}^n a_i^2} \times \sqrt{\sum_{i=1}^n b_i^2}}$$

Figure 6. *Cosine Similarity Between Nonzero Vectors a and b in n -Dimensional Space*

The application of cosine similarity to candidate-job matching involves computing the similarity between each applicant's embedding and each job's embedding, producing a similarity score for each applicant-job pair. Armansyah et al. (2026) examine threshold-based application, where a similarity threshold is used to filter or flag applicant-job pairs for further review. Holis et al. (2025) apply semantic matching to FAQ chatbot evaluation, demonstrating the breadth of cosine similarity's application. For the proposed FIRA platform, cosine similarity is applied after SBERT embedding generation, producing a similarity score for each applicant-job pair that serves as an input to the LightGBM ranking component. The raw cosine similarity score is retained for ranking and audit purposes, while a nonnegative similarity index may be presented as a display convention. Clipping negative values creates display ties, so candidates are ranked using the raw score rather than the clipped index.

Figure 7. *Pseudocode for Cosine Similarity Score Calculation*

```

FUNCTION cosine_similarity(vec_a, vec_b):
    // Both vectors must be nonzero and same dimensionality
    IF length(vec_a) != length(vec_b) THEN
        RETURN ERROR("Dimension mismatch")
    END IF
    // Compute the dot product
    dot_product <- 0
    FOR i FROM 1 TO length(vec_a) DO
        dot_product <- dot_product + (vec_a[i] * vec_b[i])
    END FOR
    // Compute L2 norms of each vector
    norm_a <- sqrt(sum(vec_a[i] * vec_a[i] for i in 1..length(vec_a)))
    norm_b <- sqrt(sum(vec_b[i] * vec_b[i] for i in 1..length(vec_b)))
    // Handle zero-vector edge case
    IF norm_a == 0 OR norm_b == 0 THEN
        RETURN 0 // Cannot compute angle for zero vector
    END IF
    similarity <- dot_product / (norm_a * norm_b)
    RETURN similarity // Value in [-1, 1]
END FUNCTION

```

Previous applications of cosine similarity to candidate-job matching and information retrieval establish it as the standard comparison measure for textual embeddings. Manning, Raghavan, and Schütze (2008) establish cosine similarity as a foundational measure in information retrieval, where it is used to rank documents against queries represented as term vectors. Reimers and Gurevych (2019) apply cosine similarity to compare SBERT-generated sentence embeddings on the Sentences Involving Compositional Knowledge benchmark, demonstrating that cosine similarity on SBERT embeddings outperforms BERT's cross-encoder on speed while retaining strong accuracy. Rieon et al. (2026) apply cosine similarity to SBERT embeddings in a recruitment matching context and report strong retrieval performance. Bevara et al. (2025) apply cosine similarity to resume embeddings in the Resume2Vec system, transforming applicant tracking through semantic representation. These prior

applications confirm that cosine similarity is a validated comparison measure for the semantic component of the proposed FIRA platform's matching pipeline.

The advantages of cosine similarity include its simplicity, low computational cost for normalized embeddings where the dot product equals the cosine similarity, bounded range that supports interpretation, and scale-invariance that prevents magnitude differences from dominating the comparison. The limitations of cosine similarity include its sensitivity to the quality of the embeddings, the lack of a universal interpretation of a particular score for hiring decisions, and the dependence on the embedding model's training data and architecture. Steck et al. (2024) note that the meaning of the score depends on how representations are constructed, and that the absolute value is not a probability of qualification. For the proposed FIRA platform, these limitations are addressed through the integration of cosine similarity with structured features in the LightGBM ranking component, which produces a more nuanced ranking than cosine similarity alone, and through the matched and missing skills identification, which provides a traceable explanation for the similarity score.

LightGBM is the gradient-boosted decision-tree framework that the proposed platform adopts for ranking candidates based on integrated features. Ke et al. (2017) introduced LightGBM, describing techniques for efficient training including Gradient-based One-Side Sampling and Exclusive Feature Bundling. Li (2024) applied LightGBM to enterprise recruitment and reported strong performance with efficient training, with results of 78.6 percent accuracy, 75.4 percent precision, 80.1 percent recall, and 77.7 percent F1 score in recruitment matching. Unlike SBERT, LightGBM does not directly supply a semantic representation of text; its role is to combine

already-constructed features, such as semantic similarity and structured requirement comparisons, within a supervised prediction or ranking task.

LightGBM is a gradient-boosted decision-tree framework designed to learn predictive relationships from feature data. Ke et al. (2017) describe the core techniques: leaf-wise tree growth, which grows the tree by splitting the leaf with the highest delta loss rather than level by level, achieving lower loss for a given number of leaves; histogram-based splitting, which bins continuous features into discrete buckets, reducing memory usage and accelerating split finding; Gradient-based One-Side Sampling, which focuses on samples with larger gradients to reduce training cost; and Exclusive Feature Bundling, which bundles mutually exclusive features to reduce dimensionality. Li (2024) found that these techniques make LightGBM faster than XGBoost while achieving comparable or better accuracy. The proposed FIRA platform adopts the ranking objective, where LightGBM produces scores used to order candidates within a job group rather than to estimate a probability of qualification.

Gradient boosting decision trees build an ensemble sequentially, with each tree correcting the errors of the current ensemble. Ke et al. (2017) describe the process: an initial prediction is made, the gradient of the loss function is computed, and each subsequent tree is fit to the negative gradient, with a learning rate controlling the contribution of each tree. The additive model expresses the ensemble score as the initial prediction plus the learning rate times the sum of tree contributions, as shown in Figure 8. For ranking applications, labels and job-group boundaries are essential because the learning task concerns relative relevance among candidates for the same vacancy. A classifier, regressor, and ranker are different modeling choices, and calling

the output a match probability without specifying and validating a probabilistic classification task would be incorrect. The proposed FIRA platform adopts the ranking objective, where LightGBM produces scores used to order candidates within a job group.

$$F(x) = F_0(x) + \eta \sum_{t=1}^T f_t(x)$$

Figure 8. *LightGBM Additive Ensemble Model for Feature Vector x*

where $F(x)$ is the ensemble score for feature vector x , $F_0(x)$ is the initial prediction, $f_t(x)$ is the contribution of tree t , η is the learning rate, and T is the number of trees. The leaf-wise growth strategy selects the leaf with the highest delta loss for splitting, which produces deeper, more asymmetric trees than the level-wise growth strategy used by traditional gradient boosting. This strategy achieves lower loss for a given number of leaves, but requires careful control of tree complexity to prevent overfitting. Figure 9 presents the LightGBM leaf-wise growth and prediction process in pseudocode form.

Figure 9. *Pseudocode for LightGBM Training and Leaf-Wise Tree Growth*

```

FUNCTION train_lightgbm(training_data, labels, params):
    // Initialize the model with the first prediction
    F_0 <- compute_initial_prediction(labels)
    ensemble <- [] // Empty list of trees
    residuals <- labels - F_0
    FOR t FROM 1 TO params.num_trees DO
        // GOSS: sample instances with larger gradients more often
        sampled <- gradient_based_one_side_sampling(residuals, params)
        // EFB: bundle mutually exclusive features to reduce dimensionality
        bundled <- exclusive_feature_bundling(sampled, params)
        // Build a single tree using leaf-wise (best-first) growth
        tree <- build_tree_leaf_wise(bundled, residuals, params)
        // Add the new tree to the ensemble
        APPEND tree TO ensemble
        // Update residuals using the learning rate
        predictions <- F_0 + params.eta * sum(tree.predict(x) for tree in
ensemble)
        residuals <- labels - predictions
        // Early stopping check on validation set
        IF validation_loss_not_improving(patience) THEN BREAK
        END IF
    END FOR
    RETURN ensemble
END FUNCTION

FUNCTION build_tree_leaf_wise(data, gradients, params):
    root <- create_leaf(data, gradients)
    leaves <- priority_queue_by_delta_loss()
    PUSH root INTO leaves
    WHILE size(leaves) > 0 AND num_leaves(tree) < params.max_leaves DO
        best_leaf <- POP(leaves)
        // Histogram-based split finding on the best leaf
        split <- find_best_split(best_leaf, params)
        IF split.gain <= params.min_split_gain THEN CONTINUE
        END IF
        left, right <- apply_split(best_leaf, split)
        PUSH left INTO leaves
        PUSH right INTO leaves
    END WHILE
    RETURN tree
END FUNCTION

```

Feature engineering for candidate-job matching involves constructing the features that LightGBM uses for ranking. Jirjees (2025) and Li (2024) describe features including experience years, skill-match ratio, and education alignment. Sugiura (2024) and Zhao et al. (2024) describe user-item features, behavioral features, contextual features, and content features. For the proposed FIRA platform, features include the cosine similarity score between applicant and job embeddings, documented skill overlap (the intersection of skills listed in the applicant's profile and required in the job description), experience satisfaction (whether the applicant's years of experience meet or exceed the job's requirement), document completeness (whether the applicant has submitted all required documents), and categorical descriptors such as education level and certification status. These features are integrated into a feature vector for each applicant-job pair, which LightGBM uses to produce the ranked output. Missing data is distinguished from a verified absence of a qualification, as the distinction affects the ranking and the interpretation of the results.

Feature selection and importance analysis identify which features contribute most to the ranking and support the interpretability of the model. Jamal (2024) applied SHAP, LIME, and permutation feature importance to recruitment data and found that feature importance analysis increases recruiter trust and reduces time-to-decision. Talebi and Khatibi Bardsiri (2025) combined genetic algorithms with LightGBM for feature selection in employee turnover prediction. Nowak and Pawłowska-Nowak (2026) applied SHAP-based feature contributions in an interpretable multilayered recruitment model. For the proposed FIRA platform, feature importance analysis serves two purposes: it identifies which features drive the ranking, supporting the transparency of the matching

results, and it guides feature engineering by highlighting which features contribute to matching accuracy. However, a global feature-importance chart summarizes model behavior across data and does not explain every individual recommendation. Local explanations may clarify how the model used particular features for a specific candidate, but they do not prove fairness or credential validity.

Model training and prediction involve fitting the LightGBM model to labeled training data and using the trained model to rank candidates for new applicant-job pairs. Li (2024) provides benchmark training results for LightGBM in recruitment matching. Kannen et al. (2024) examined pointwise and pairwise learning-to-rank objectives, which determine how the model is trained to produce rankings. For the proposed FIRA platform, training data consists of applicant-job pairs with relevance labels, where the labels indicate the quality of the match based on recruitment outcomes or human-reviewed relevance judgments. Historical acceptance decisions should not automatically become ground truth, as decisions may reflect unrecorded constraints or previous bias, and the concerns identified by Fabris et al. (2025) apply to the training labels as well as to protected attributes in the input. Prediction involves applying the trained model to new applicant-job pairs, producing a ranking score for each pair that is used to order candidates for the vacancy.

Hyperparameter tuning optimizes the LightGBM model's configuration to achieve the best performance on held-out data. Xiong et al. (2024) applied Optuna with Tree-structured Parzen Estimator Bayesian optimization for LightGBM hyperparameter tuning. Ying et al. (2025) and Talebi et al. (2025) provide additional evidence on systematic hyperparameter optimization. Al Akasheh et al. (2024) conducted a

systematic review of boosting approaches and their hyperparameter sensitivity. The key hyperparameters for LightGBM include the number of trees, learning rate, maximum tree depth, number of leaves, minimum data per leaf, and feature fraction. These hyperparameters control the model's complexity and its tendency to overfit. Early stopping, where training halts when performance on a validation set stops improving, prevents overfitting by limiting the number of trees to the point of optimal generalization. The proposed FIRA platform adopts bounded hyperparameter configurations with early stopping, learning rate discipline, and leaf count constraints, with the specific configuration documented as part of the matching method and evaluated through cross-validation.

Previous applications of LightGBM in ranking and recruitment contexts establish its fitness for the candidate-ranking component of the proposed platform. Ke et al. (2017) introduced LightGBM and demonstrated state-of-the-art performance on standard machine learning benchmarks with significantly lower training time than competing gradient-boosting frameworks. Li (2024) applied LightGBM to enterprise recruitment matching and reported strong ranking performance with efficient training characteristics that suit production deployment. Jirjees (2025) applied gradient-boosting frameworks including LightGBM to candidate-job matching tasks, finding that feature integration with semantic embeddings produces stronger rankings than semantic similarity alone. Talebi and Khatibi Bardsiri (2025) combined genetic algorithms with LightGBM for feature selection in employee turnover prediction, demonstrating LightGBM's capacity to handle human resource data with mixed numerical and categorical feature types. These prior

applications confirm that LightGBM is a validated ranking model for the predictive component of the proposed FIRA platform's matching pipeline.

The advantages of LightGBM include its training and inference speed, its ability to handle mixed numerical and categorical feature types, its interpretability through feature importance analysis, and its efficiency through leaf-wise growth, histogram-based splitting, Gradient-based One-Side Sampling, and Exclusive Feature Bundling. Li (2024) found that LightGBM is faster than XGBoost while achieving comparable or better accuracy. The limitations of LightGBM include the risk of overfitting on small datasets, sensitivity to hyperparameter choices, opacity of individual tree decisions, and the dependence on labeled training data whose quality and representativeness affect the model's fairness and accuracy. Han et al. (2023) combined LightGBM with other components in a hybrid recommendation model, supporting the use of tree-based feature processing within a recommendation pipeline. Kapoor and Narayanan (2023) caution against data leakage in model evaluation. The proposed FIRA platform addresses these limitations through cross-validation, hyperparameter discipline, feature importance analysis for interpretability, and the decision-support framing that retains human authority over final decisions.

The integration of SBERT, cosine similarity, and LightGBM into a single pipeline is the core architectural contribution of the proposed FIRA platform's matching component. The pipeline combines the semantic understanding of SBERT with the efficient comparison of cosine similarity and the predictive ranking of LightGBM, producing ranked candidate lists with associated similarity scores and matched and missing skill identification as decision-support information. Shen et al. (2024) describe

learning-to-retrieve as the production standard for large-scale matching, where embedding retrieval, similarity scoring, and learning-to-rank are combined. The integration enables the platform to consider both the overall semantic alignment and the specific requirement satisfaction, producing a more nuanced ranking than either component alone.

The first stage of the pipeline is SBERT embedding generation, where applicant and job textual information is preprocessed and encoded into fixed-dimensional vectors. Rosenberger et al. (2025) and Rieon et al. (2026) demonstrate this stage in recruitment matching contexts. Liu et al. (2025) report that batched, GPU-accelerated SBERT embedding can process approximately 1,000 resumes per minute. For the proposed FIRA platform, this stage involves preprocessing resume and job description text, applying the SBERT model's tokenizer, encoding the text through the transformer encoder, and applying the pooling operation to produce the sentence embedding. The embeddings are normalized to unit length to facilitate cosine similarity computation. The embedding generation is independent for each applicant and job, meaning that stored embeddings can be reused when the source text has not changed, providing computational efficiency for the comparison stage.

The second stage of the pipeline is cosine similarity score calculation, where the normalized embeddings are compared to produce a similarity score for each applicant-job pair. Rieon et al. (2026) and Kurek et al. (2024) demonstrate this stage. For normalized embeddings, the cosine similarity equals the dot product, which is a single matrix multiplication operation that can be computed efficiently for all applicant-job pairs simultaneously. The raw cosine similarity score ranges from negative

one to one, and the proposed platform may present a nonnegative similarity index of 100 times the maximum of zero and the raw score as a display convention, while retaining the raw value for ranking and audit purposes. Clipping negative values creates display ties, so candidates are ranked using the raw score rather than the clipped index.

The third stage of the pipeline is feature integration, where the cosine similarity score is combined with structured features to form the feature vector that LightGBM uses for ranking. Jirjees (2025) and Li (2024) describe the integration of semantic similarity with structured features. For the proposed FIRA platform, the feature vector for each applicant-job pair includes the cosine similarity score, documented skill overlap, experience satisfaction, document completeness, and categorical descriptors such as education level and certification status. The integration of semantic and structured features enables the ranking model to consider both the overall semantic alignment and the specific requirement satisfaction, producing a more nuanced ranking than either component alone. Sugiura (2024) and Zhang et al. (2025) confirm that feature integration is a standard practice in production matching systems, and that feature importance analysis through approaches such as SHAP provides interpretable feedback on which features drove the ranking.

The fourth stage of the pipeline is LightGBM-based prediction and ranking, where the trained LightGBM model produces a ranking score for each applicant-job pair based on the integrated feature vector. Shen et al. (2024) describe learning-to-retrieve as the production standard for large-scale matching, where embedding retrieval, similarity scoring, and learning-to-rank are combined. Kannen et al. (2024) examined pointwise and pairwise learning-to-rank objectives. For the proposed FIRA platform, the LightGBM

model is trained on applicant-job pairs with relevance labels and produces a ranking score for each new pair. The scores are used to order candidates within a job group, producing the ranked candidate list that is presented as decision-support information to recruitment personnel and employers. The ranking is advisory, and the final selection decision remains with authorized human users who can evaluate factors not captured in the feature vector. Figure 10 presents the integrated FIRA matching pipeline in pseudocode form.

Figure 10. *Pseudocode for the Integrated FIRA Matching Pipeline*

INPUT: applicant_pool, job, sbert_model, lightgbm_model
OUTPUT: ranked_candidate_list, matched_missing_skills_report

FUNCTION match_candidates(applicant_pool, job):

// Stage 1: SBERT embedding generation for the job

job_text <- concatenate(job.title, job.description, job.skills, job.requirements)

job_embedding <- generate_embedding(job_text, sbert_model)

candidates <- []

// Process each applicant in the pool

FOR EACH applicant **IN** applicant_pool **DO**

// Stage 1: SBERT embedding generation for the applicant

applicant_text <- concatenate(applicant.resume, applicant.profile,
applicant.skills, applicant.qualifications)

applicant_embedding <- generate_embedding(applicant_text,
sbert_model)

// Stage 2: Cosine similarity score calculation

cosine_score <- cosine_similarity(applicant_embedding, job_embedding)

// Stage 3: Feature integration

skill_overlap <- compute_skill_overlap(applicant.skills, job.required_skills)

experience_match <- check_experience(applicant.years_exp,
job.min_years_exp)

document_complete <- check_documents(applicant.documents,
job.required_documents)

feature_vector <- [cosine_score, skill_overlap, experience_match,
document_complete, applicant.education_level,
applicant.certification_status]

// Stage 4: LightGBM prediction

rank_score <- lightgbm_model.predict(feature_vector)

// Identify matched and missing skills for transparency

matched, missing <- identify_matched_missing_skills(applicant.skills,
job.required_skills)

APPEND {applicant, cosine_score, rank_score, matched, missing} **TO**
candidates

END FOR

// Stage 5: Rank candidates by LightGBM score, descending

ranked_list <- sort(candidates, by=rank_score, descending=TRUE)

RETURN ranked_list, extract_matched_missing_report(ranked_list)
END FUNCTION

The final candidate-job matching score is the output of the pipeline, presented as decision-support information to recruitment personnel and employers. The score combines the LightGBM ranking score with the cosine similarity score and the matched and missing skills identification, providing a comprehensive view of the applicant-job alignment. The percentage similarity score is presented as an advisory index rather than a probability, consistent with the caveat that raw cosine similarity is not a probability of qualification (Steck et al., 2024; Muennighoff et al., 2023). The matched and missing skills identification provides a traceable basis for each match, showing which required skills the applicant possesses and which are absent, supporting the recruiter's ability to understand, trust, and challenge the matching results. The final output is a ranked candidate list with associated scores and skill analysis, presented as decision-support information that supports rather than replaces human evaluation, consistent with the decision-support framing established in Chapter 1 and the ethical requirements identified in the reviewed literature.

The data sets used for training and evaluating the matching component include synthetic data, raw data from recruitment records, and annotated data with relevance labels. The quality, representativeness, and proper handling of these data sets are critical to the fairness, accuracy, and generalizability of the matching results. Foronda (2023) examined data handling in Philippine institutions subject to the Data Privacy Act and emphasized that raw data must be handled in compliance with consent, purpose limitation, data minimization, and security requirements.

Synthetic data may be used to exercise the software pipeline and test the matching component's behavior under controlled conditions. Long et al. (2024) describe over 300

synthetic datasets available on Hugging Face for testing machine learning pipelines. Nadaş (2025) cautions that synthetic data generated by large language models may not generalize to real recruitment contexts, as the distribution of skills, experience, and qualifications in synthetic data may differ from the distribution in actual applicant pools. For the proposed FIRA platform, synthetic data may be used for pipeline testing and development, but it cannot substitute for independent evaluation on realistic, human-reviewed applicant-job pairs. The platform's evaluation distinguishes between synthetic-data testing, which validates the pipeline's technical functionality, and real-data evaluation, which validates the matching component's effectiveness in the actual recruitment context.

Raw data for the matching component includes resumes, applicant profiles, job descriptions, skills, qualification requirements, and recruitment outcomes. Foronda (2023) emphasized that raw data must be handled in compliance with consent, purpose limitation, data minimization, and security requirements. For the proposed FIRA platform, raw data is heterogeneous, comprising free-text resumes, structured qualification fields, job descriptions with varying formats, and application status histories. The data is subject to the Data Privacy Act of 2012, which mandates role-based access control, encryption, audit logging, and data subject rights. The raw data is preprocessed to extract the textual information that feeds the SBERT embedding generation, with the preprocessing approach documented as part of the matching method.

The data annotation and labeling process assigns relevance labels to applicant-job pairs, indicating the quality of the match based on recruitment outcomes or

human-reviewed relevance judgments. Artemova et al. (2024) describe hybrid labeling, where a language model proposes labels and a human reviewer validates them, providing a scalable approach to annotation. For the proposed FIRA platform, match-quality labels are based on recruitment outcomes, such as whether the applicant was endorsed, selected, and successfully deployed, and on human-reviewed relevance judgments by authorized agency personnel. Historical acceptance decisions should not automatically become ground truth, as decisions may reflect unrecorded constraints or previous bias, and the concerns identified by Fabris et al. (2025) apply to the training labels. The labeling process is documented as part of the matching method, with attention to the sources of labels, the potential for bias, and the distinction between label quality and the quality of the underlying match.

Model training and validation encompass the processes of fitting the LightGBM model to labeled data, evaluating its performance on held-out data, and preventing overfitting to ensure that the model generalizes to new applicant-job pairs. Eltahier et al. (2025) and Conciatori et al. (2024) describe the partitioning of data into training and testing sets. For the proposed FIRA platform, the data is partitioned such that the testing set reflects the intended deployment question, namely performance on previously unseen job groups and applicants, rather than random row-level splitting. Kapoor and Narayanan (2023) caution that random splitting can place closely related information in both training and test sets, causing data leakage that inflates performance estimates. The same applicant may appear in multiple applications, and several candidates may share one job description, requiring group-aware splitting that accounts for these dependencies.

Cross-validation is the process of partitioning the data into multiple folds, training on a subset, and evaluating on the remaining fold, repeated across all folds to produce a robust performance estimate. Abedin et al. (2026) describe k-fold cross-validation with k equal to 5 or 10 as a standard approach. Lumumba et al. (2024) recommend repeated k-fold cross-validation for imbalanced data, where the positive class of qualified applicants is rare, as repeated cross-validation provides a more stable performance estimate than a single k-fold run. Nowak and Pawłowska-Nowak (2026) applied cross-validation in an end-to-end recruitment matching evaluation. Kapoor and Narayanan (2023) emphasize that cross-validation must account for group structure to prevent leakage. For the proposed FIRA platform, the cross-validation approach combines repeated, stratified, group-aware k-fold, which addresses the imbalanced nature of recruitment data, the group structure of applicant-job pairs, and the need for stable performance estimates.

Overfitting prevention ensures that the model generalizes to new data rather than memorizing the training set. Abedin et al. (2026) describe the bias-variance trade-off in k-fold cross-validation. Ying et al. (2025) and Al Akasheh et al. (2024) describe architectural precautions including tree complexity control, minimum data per leaf, learning rate discipline, and training duration control. Lumumba et al. (2024) recommend monitoring the validation gap during training to detect overfitting. For the proposed FIRA platform, overfitting is prevented through regularization, cross-validation, held-out test sets, early stopping, and architectural restraint. The specific overfitting prevention measures are documented as part of the matching method, and the model's

generalization is evaluated on the held-out test set to provide an unbiased estimate of performance.

Model performance is evaluated using metrics that capture the matching component's effectiveness in ranking candidates. Li (2024) reports LightGBM performance in recruitment matching of 78.6 percent accuracy, 75.4 percent precision, 80.1 percent recall, and 77.7 percent F1 score. Rainio et al. (2024) caution that single-metric reporting can be misleading, and recommend reporting multiple metrics together. Sujon et al. (2025) found that F1 is the most stable metric under class imbalance. Conciatori et al. (2024) recommend reporting accuracy, precision, recall, and F1 together with a confusion matrix. Holis et al. (2025) report F1 scores of approximately 0.79 to 0.96 in related matching tasks. Pin (2026b) reports industry F1 of approximately 0.91. For the proposed FIRA platform, performance is evaluated using accuracy, precision, recall, F1 score, and, where appropriate, information retrieval ranking metrics such as precision at k, recall at k, and normalized discounted cumulative gain, with the specific metrics documented as part of the evaluation method.

The scope and limitations of the matching component are defined by the constraints of the approach and the data. The matching component is limited to the textual information available from applicant profiles, resumes, job descriptions, skills, and qualification requirements, and cannot evaluate factors such as interview performance, interpersonal abilities, personality, work attitude, and other competencies not represented in the available text, as established in Chapter 1. The matching results are dependent on the selected SBERT model, the preprocessing approach, and the information provided to the system, and different models, datasets, preprocessing

techniques, or similarity approaches may produce different results. Michail and Clematide (2025) found cross-lingual variance in performance, affecting the platform's effectiveness on Filipino text. Fabris et al. (2025) identify measurement, representation, and aggregation bias as failure modes in algorithmic hiring, establishing that matching results can reflect and amplify existing biases in the training data. These limitations are addressed through the decision-support framing, which retains human authority over final decisions, through transparency features that make the basis for each match reviewable, and through ongoing evaluation and documentation of the matching component's scope and limitations.

EVALUATION

The evaluation of the proposed FIRA platform encompasses five dimensions: software quality, usability, performance and load behavior, machine learning matching accuracy, and ethical and legal compliance. The International Organization for Standardization (2023) established ISO/IEC 25010 as the framework for software quality evaluation, with the 2023 edition superseding the 2011 edition. The evaluation framework draws on multiple standards and instruments, including ISO/IEC 25010 for software quality characteristics, the System Usability Scale for usability assessment, the Technology Acceptance Model for user acceptance, Apache JMeter for performance and load testing, and precision, recall, and F1 score for matching accuracy. Khan et al. (2025) conducted a confirmatory factor analysis of the System Usability Scale, and Almeida (2025) applied the Technology Acceptance Model to AI recruitment contexts.

The evaluation involves both IT experts, who assess the technical and functional characteristics of the developed system, and intended end users, including applicants, FIRA staff, and employers or principals, who assess usability, accessibility, ease of use, and the appropriateness of the system for recruitment-related activities. The candidate-job matching component is assessed based on its generated similarity scores and candidate rankings to determine its usefulness as decision-support information. This dual-evaluation approach, combining expert and end-user perspectives, ensures that both technical and experiential dimensions of system quality are captured.

Apache JMeter is the performance testing tool the proposed platform adopts for load and stress testing. The Apache Software Foundation (2024) maintains JMeter as an open-source tool for load and performance testing of web applications. Annisa et al. (2026) describe web application performance testing using Apache JMeter, and Bolanowski et al. (2024) propose a layered performance test model for web applications. Analytical Evaluation (2023) compares Apache JMeter and SoapUI as web performance testing tools. Performance testing identifies the platform's response time, resource utilization, and system capacity under varying workloads, ensuring that the platform can handle the volume of applicants and job postings that a Philippine land-based OFW recruitment agency typically processes.

Response time measures the time the platform takes to respond to user requests, with target thresholds established based on user expectations and industry benchmarks. Resource utilization measures the platform's consumption of server resources, including CPU, memory, and network bandwidth, under varying workloads. System capacity measures the maximum number of concurrent users and requests that

the platform can handle while maintaining acceptable response times. The Apache Software Foundation (2024) provides JMeter user manual guidance on configuring load tests, ramp-up periods, and thread groups to simulate realistic user behavior. For the proposed FIRA platform, performance testing scenarios are based on the agency's actual use patterns rather than borrowed from large recruitment platform benchmarks, ensuring that the test results reflect the platform's expected operational context.

ISO/IEC 25010 is the international standard for software quality evaluation, providing a framework of quality characteristics that the proposed platform uses to guide its evaluation. The International Organization for Standardization (2023) issued the 2023 edition of ISO/IEC 25010, which supersedes the 2011 edition. Rafi et al. (2025) and Rojas et al. (2025) apply the ISO/IEC 25010 framework to recruitment system evaluation, with Rojas et al. (2025) conducting a bibliometric analysis of 1,613 publications on software quality. The framework identifies eight quality characteristics: functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability. The proposed FIRA platform focuses its evaluation on four characteristics most relevant to its operational context: functional suitability, usability, performance efficiency, and reliability.

Functional suitability assesses whether the software meets its intended requirements in terms of completeness, correctness, and appropriateness. For the proposed FIRA platform, functional suitability is evaluated through functional testing that verifies each feature against its specification, including applicant registration, job posting, application submission, status tracking, matching, endorsement, and reporting. Pacific Certifications (2025) and ISO 25010 (n.d.) provide guidance on functional

suitability evaluation. The evaluation involves both expert review and end-user testing, with defects recorded and tracked through resolution.

Usability assesses the degree to which the software can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use. For the proposed FIRA platform, usability is evaluated through the System Usability Scale (SUS) administered to representative end users, including applicants, FIRA staff, and employers. Brooke (1996) introduced SUS as a quick and dirty usability scale, and Khan et al. (2025) conducted a confirmatory factor analysis validating its use. The usability evaluation captures both quantitative SUS scores and qualitative feedback, providing actionable information for interface refinement.

Performance efficiency assesses the performance of the software relative to the resources used under stated conditions. For the proposed FIRA platform, performance efficiency is evaluated through JMeter load and stress testing, measuring response time, resource utilization, and system capacity under varying workloads. The performance evaluation identifies the platform's capacity to handle growth in applicant pools, concurrent users, and matching workloads, informing decisions about infrastructure scaling. Reliability assesses the degree to which the software performs its specified functions under specified conditions for a specified period. For the proposed FIRA platform, reliability is evaluated through stress testing that identifies the conditions under which the platform fails, and through monitoring of error rates and recovery times.

Statistical analysis of evaluation results provides the basis for interpreting the platform's performance and identifying areas for improvement. The evaluation framework draws on multiple metrics, including accuracy, precision, recall, F1 score,

and information retrieval ranking metrics such as precision at k, recall at k, and normalized discounted cumulative gain. Rainio et al. (2024) caution that single-metric reporting can be misleading, and recommend reporting multiple metrics together. Conciatori et al. (2024) recommend reporting accuracy, precision, recall, and F1 together with a confusion matrix. The platform's evaluation reports these metrics together with confidence intervals where appropriate, providing a comprehensive view of matching accuracy.

The Technology Acceptance Model (TAM) provides a framework for evaluating user acceptance of the platform. Davis (1989) introduced TAM, which models user acceptance as a function of perceived usefulness and perceived ease of use. Almeida (2025) applied TAM to AI-assisted recruitment contexts, and Akram (2024) extended TAM for human-centered AI systems. Alshammari and Al-Mamary (2025) extended TAM to AI-powered training systems. For the proposed FIRA platform, TAM is administered to representative end users to capture perceived usefulness, perceived ease of use, attitude toward use, and behavioral intention to use. The TAM results complement the SUS results, providing a multidimensional view of user acceptance.

User acceptance and feedback are collected through structured surveys and interviews with representative end users, including applicants, FIRA staff, and employers. Enad and Pabalan (2023) and Enad (2025) provide local Philippine context for recruitment system user acceptance evaluation. The feedback collection captures both quantitative ratings and qualitative comments, providing actionable information for platform refinement. The evaluation also captures demographic and contextual

information, allowing analysis of how user characteristics affect acceptance and usability perceptions.

Training parameters and model configuration for the matching component are documented as part of the evaluation method, ensuring reproducibility and enabling comparison with future iterations. Eltahier et al. (2025) and Conciatori et al. (2024) describe the importance of documenting training parameters for machine learning model evaluation. For the proposed FIRA platform, training parameters include the selected SBERT model checkpoint, fine-tuning configuration, LightGBM hyperparameters, cross-validation configuration, and early stopping criteria. These parameters are documented in a configuration file that is versioned alongside the model, ensuring that the evaluation results can be reproduced and compared with future iterations.

Precision, recall, and F1-score are the standard metrics for evaluating matching accuracy in classification and ranking tasks. Shaukat (2023) describes the use of these metrics in machine learning model evaluation. Accuracy measures the proportion of correctly predicted outcomes over the total predictions. Precision indicates how many of the predicted positive cases are actually correct. Recall, also known as sensitivity, measures the ability of the system to identify all relevant cases within a dataset. The F1-score, which balances both metrics, is the harmonic mean of precision and recall. Together, these measures provide a robust framework for evaluating the predictive performance of algorithms in diverse applications. Figure 11 presents the formulas for these metrics.

Figure 11. *Algorithm Performance Metrics Formulas for Accuracy, Precision, Recall, and F1-Score*

Metric	Formula	Interpretation
Accuracy	$(TP + TN) / (TP + TN + FP + FN)$	Overall correctness of the model
Precision	$TP / (TP + FP)$	Reliability of positive predictions
Recall	$TP / (TP + FN)$	Coverage of actual positive cases
F1-Score	$2TP / (2TP + FP + FN)$	Harmonic mean of precision and recall

where TP is true positives, TN is true negatives, FP is false positives, and FN is false negatives. For the candidate-job matching task, a true positive is an applicant-job pair that the model correctly identifies as a good match, a true negative is an applicant-job pair that the model correctly identifies as a poor match, a false positive is an applicant-job pair that the model incorrectly identifies as a good match, and a false negative is an applicant-job pair that the model incorrectly identifies as a poor match. The choice of threshold for classifying a pair as a good match affects the trade-off between precision and recall, with higher thresholds producing higher precision at the cost of lower recall, and lower thresholds producing higher recall at the cost of lower precision.

The matching component's performance is compared against two baselines: manual matching and keyword-based matching. Manual matching, the traditional approach, involves human recruiters reviewing each applicant against the job requirements, producing a subjective and unscalable ranking. Ajjam et al. (2025)

demonstrated that semantic similarity matching outperforms keyword-based comparison on standard retrieval metrics. Kurek et al. (2024) found that an SBERT-family model surfaced the correct candidate within the top-500 results far more reliably than a keyword baseline. Aricheta, Santiano, and Ayo (2026) frame the Philippine online job market as suffering from information overload and a heavy reliance on keyword-based search mechanisms that routinely fail to capture semantic skill alignment. The comparison against these baselines establishes whether the proposed SBERT-based semantic matching contributes beyond shared vocabulary and reduces the cognitive burden on human recruiters.

Cross-validation and overfitting checks ensure that the matching component generalizes to new applicant-job pairs rather than memorizing the training set. Abedin et al. (2026) describe k-fold cross-validation with k equal to 5 or 10 as a standard approach. Lumumba et al. (2024) recommend repeated k-fold cross-validation for imbalanced data. Kapoor and Narayanan (2023) emphasize that cross-validation must account for group structure to prevent leakage. For the proposed FIRA platform, the cross-validation approach combines repeated, stratified, group-aware k-fold, which addresses the imbalanced nature of recruitment data, the group structure of applicant-job pairs, and the need for stable performance estimates. Model generalization is evaluated on the held-out test set to provide an unbiased estimate of performance, and the validation gap during training is monitored to detect overfitting.

Scalability and load testing identify the platform's capacity to handle growth in applicant pools, concurrent users, and matching workloads. Bolanowski et al. (2024) propose a layered performance test model for web applications. APMdigest (2024)

describes scalability, load, and stress testing in web applications, distinguishing between load testing (expected load), stress testing (beyond-design load), and scalability testing (growth over time). For the proposed FIRA platform, scalability and load testing are conducted using Apache JMeter, with workload scenarios based on the agency's actual use patterns rather than borrowed from large recruitment platform benchmarks. The testing identifies the platform's capacity to handle growth in applicant pools, concurrent users, and matching workloads, informing decisions about infrastructure scaling and the platform's ability to support future operational expansion.

Ethical and legal compliance evaluation assesses the platform's adherence to data protection, fairness, human oversight, and recruitment policy requirements. Rigotti and Fosch-Villaronga (2024) emphasize the need for accountability and the right to challenge automated decisions in AI-based recruitment. Fabris et al. (2025) catalogued bias sources and fairness requirements. Lazaros et al. (2026) taxonomize human-in-the-loop patterns. The evaluation framework draws on multiple standards and regulations, including the Philippine Data Privacy Act of 2012, the National Privacy Commission's 2024 AI Advisory, the European Union AI Act of 2024, the NIST AI Risk Management Framework of 2023, and the OECD AI Principles of 2024.

Data privacy compliance is assessed against the obligations established by the Data Privacy Act of 2012 and the National Privacy Commission's 2024 AI Advisory. Foronda (2023) found that consent administration and records management are the most-failed compliance areas in Philippine institutions. The evaluation assesses the platform's implementation of role-based access control, encryption, audit logging, data subject rights, and data minimization, with attention to the Section 16 right to be

informed, the right to object, and the right to erasure or rectification. Fairness in automated matching is assessed against the bias sources identified by Fabris et al. (2025), with attention to measurement bias, representation bias, and aggregation bias. The evaluation includes bias audits that compare matching outcomes across demographic groups, where such analysis is permitted under the Data Privacy Act's restrictions on the processing of sensitive personal information.

Human oversight is assessed against the human-in-the-loop patterns taxonomized by Lazaros et al. (2026), with attention to where human judgment enters the matching pipeline, what override mechanisms are available to authorized personnel, and how the platform's decision-support framing is implemented in practice. Recruitment policy compliance is evaluated through the platform's alignment with DMW requirements, its support for the five-stage application workflow, and its documentation of recruitment activities in a form that supports compliance audits. The platform's contribution to recruitment policy compliance includes providing a verifiable, regulated alternative to the informal social-media-based channels where illegal recruitment flourishes, consistent with the state policy established by Republic Act No. 8042 and its amendment, Republic Act No. 10022. The Department of Migrant Workers (2026) and Cabato and Magsambol (2025) document the regulatory context, while the World Bank (2024) provides the broader policy context.

SYNTHESIS

The reviewed literature establishes the Philippine overseas employment context as a significant, multifaceted domain that frames the need for a centralized, technology-assisted recruitment platform. The Philippine Statistics Authority (2025) estimates 2.19 million Overseas Filipino Workers in 2024, with cash remittances reaching approximately 262 billion pesos (Talabong, 2025). The Department of Migrant Workers (2026) reports the removal of tens of thousands of illegal recruitment posts from social media platforms, while Cabato and Magsambol (2025) document the escalation of illegal recruitment from 614 social media pages shut down in 2023 to over 70,000 fake online job offers thwarted in 2025. ABS-CBN News (2025) reports the closure of 32 illegal recruitment agencies in 2025 alone. Respicio and Co. (n.d.) and Cal International Philippines (n.d.) document that licensed agencies reconcile fragmented workflows across paper records, spreadsheets, email, and social media, producing information gaps that propagate throughout the recruitment process. Koman et al. (2024) and Madanchian and Taherdoost (2026) confirm that web-based recruitment has improved advertising and application capture but that integration with structured screening and decision support remains uneven. Collectively, these findings establish that the recruitment problem the proposed FIRA platform addresses is real, persistent, and shared across the Philippine land-based OFW recruitment sector, and that the platform's centralized recruitment management variable responds directly to the fragmentation that the literature identifies as the root cause of operational inefficiency.

The technologies and algorithms reviewed in the literature provide the components that the proposed FIRA platform integrates. Reimers and Gurevych (2019) introduced

SBERT as a modification of BERT (Devlin et al., 2018) that produces semantically meaningful sentence embeddings suitable for efficient comparison through cosine similarity (Manning, Raghavan, and Schütze, 2008). Ke et al. (2017) introduced LightGBM as a gradient-boosted decision-tree framework with efficient training techniques. Ajjam et al. (2025), Kurek et al. (2024), Rosenberger et al. (2025), and Rieon et al. (2026) demonstrate the application of SBERT-based semantic matching to recruitment contexts, consistently outperforming keyword-based approaches. Li (2024) and Jirjees (2025) apply LightGBM to candidate-job ranking, reporting strong performance with efficient training. Bevara et al. (2025), in developing Resume2Vec, emphasize that transforming applicant tracking requires rethinking how applicant data is structured and compared. Shen et al. (2024) describe learning-to-retrieve as the production standard for large-scale matching, where embedding retrieval, similarity scoring, and learning-to-rank are combined. Thus, the literature establishes that the algorithmic components the proposed FIRA platform adopts, namely SBERT for embedding generation, cosine similarity for similarity scoring, and LightGBM for ranking, are well-validated in recruitment and related matching contexts, and that the platform's semantic candidate-job matching variable is grounded in established algorithmic foundations.

Existing systems and platforms reviewed in the literature demonstrate the range of recruitment technologies currently available, while also revealing the absence of a platform that fits the regulated, agency-mediated, cross-border context in which FIRA operates. JobStreet (2024), LinkedIn (2024, 2025), Indeed (2026), and Workday (2024) represent the dominant commercial platforms, each built for mass-market or enterprise

hiring with keyword-based or AI-assisted matching and direct employer-to-applicant channels. Greenhouse (2025), Lever (2026), and Workable (n.d.) represent leading applicant tracking systems with structured hiring features but without the DMW-regulated deployment documentation or agency-to-principal endorsement workflow that FIRA requires. Novaković et al. (2024) note that most platforms treat the agency, the employer, and the applicant as separate accounts rather than participants in a single end-to-end recruitment pipeline. Lenasri and Shekhar (2025) confirm that the effectiveness of recruitment applications depends on integration with structured screening workflows rather than isolated job advertising. Mobo (2025) found that AI adoption in Philippine human resource management remains limited to larger organizations, leaving smaller regulated agencies without a platform that fits their context. Thus, the reviewed systems establish that the proposed FIRA platform's applicant tracking variable, implemented through a five-status workflow with role-based access for applicants, local agency staff, and foreign employers, addresses a gap that no existing platform fills.

The limitations identified across the reviewed literature converge on an integration gap rather than a technology gap. The algorithmic components exist, the regulatory framework is established, and the commercial platforms provide feature breadth, but no existing system combines DMW-regulated deployment documentation, agency-to-principal endorsement workflow, role-based access for the three stakeholder categories, transparent semantic matching, and ethical and legal compliance within a single platform designed for a Philippine land-based OFW recruitment agency. The reviewed literature identifies specific limitations: keyword-based matching misses

semantically equivalent candidates (Ajjam et al., 2025; Aricheta, Santiano, and Ayo, 2026); existing platforms treat matching results as opaque recommendations rather than transparent, auditable decision-support outputs (Zhang et al., 2025); bias sources in algorithmic hiring include measurement, representation, and aggregation bias (Fabris et al., 2025); and the regulatory perimeter governing online recruitment is established by the 2023 DMW Rules and the 2024 implementation guidelines rather than by a dedicated DMW issuance on online recruitment platforms. Among the studies reviewed, limited attention was given to the integration of regulated recruitment workflow, applicant tracking, and transparent semantic matching within a single platform designed for the FIRA context. This integration gap, supported by the literature rather than asserted without evidence, is the gap the proposed FIRA platform addresses.

The proposed FIRA platform responds to the identified integration gap by combining the algorithmic components validated in the literature with the regulated workflow and role-based architecture that the FIRA context requires. The platform's centralized recruitment management variable addresses the fragmentation documented by Respicio and Co. (n.d.) and Cal International Philippines (n.d.) through a single platform that consolidates job postings, application capture, screening, endorsement, and tracking. The platform's applicant tracking variable addresses the pipeline visibility gap documented by Novaković et al. (2024) and Bevara et al. (2025) through a five-status workflow that treats the agency, the employer, and the applicant as participants in a single end-to-end pipeline. The platform's semantic candidate-job matching variable addresses the keyword-based matching limitations documented by Ajjam et al. (2025), Kurek et al. (2024), and Aricheta, Santiano, and Ayo (2026) through

the integration of SBERT, cosine similarity, and LightGBM with matched and missing skills identification that makes the basis for each match traceable. The platform's ethical and legal compliance variable addresses the bias sources documented by Fabris et al. (2025), the human oversight patterns taxonomized by Lazaros et al. (2026), and the data protection obligations established by the Data Privacy Act of 2012 and the National Privacy Commission's 2024 AI Advisory through role-based access control, encryption, audit logging, data subject rights, and the decision-support framing that retains human authority over final hiring decisions. Collectively, these responses establish that the proposed FIRA platform's contribution lies not in any single technological advance but in the integration of existing algorithmic components into a regulated, agency-specific recruitment environment that the reviewed literature identifies as lacking.

CONCEPTUAL MODEL OF THE STUDY

To illustrate the development of the system, this study uses an Input-Process-Output (IPO) model that demonstrates how the initial resources and materials feed into the system's processes, which then produce a working and evaluated web-based application. The model synthesizes the reviewed literature into a coherent representation of how knowledge, software, hardware, and data inputs are converted into recruitment management and intelligent matching outputs, with an evaluation phase that closes the loop and feeds refinements back into development.

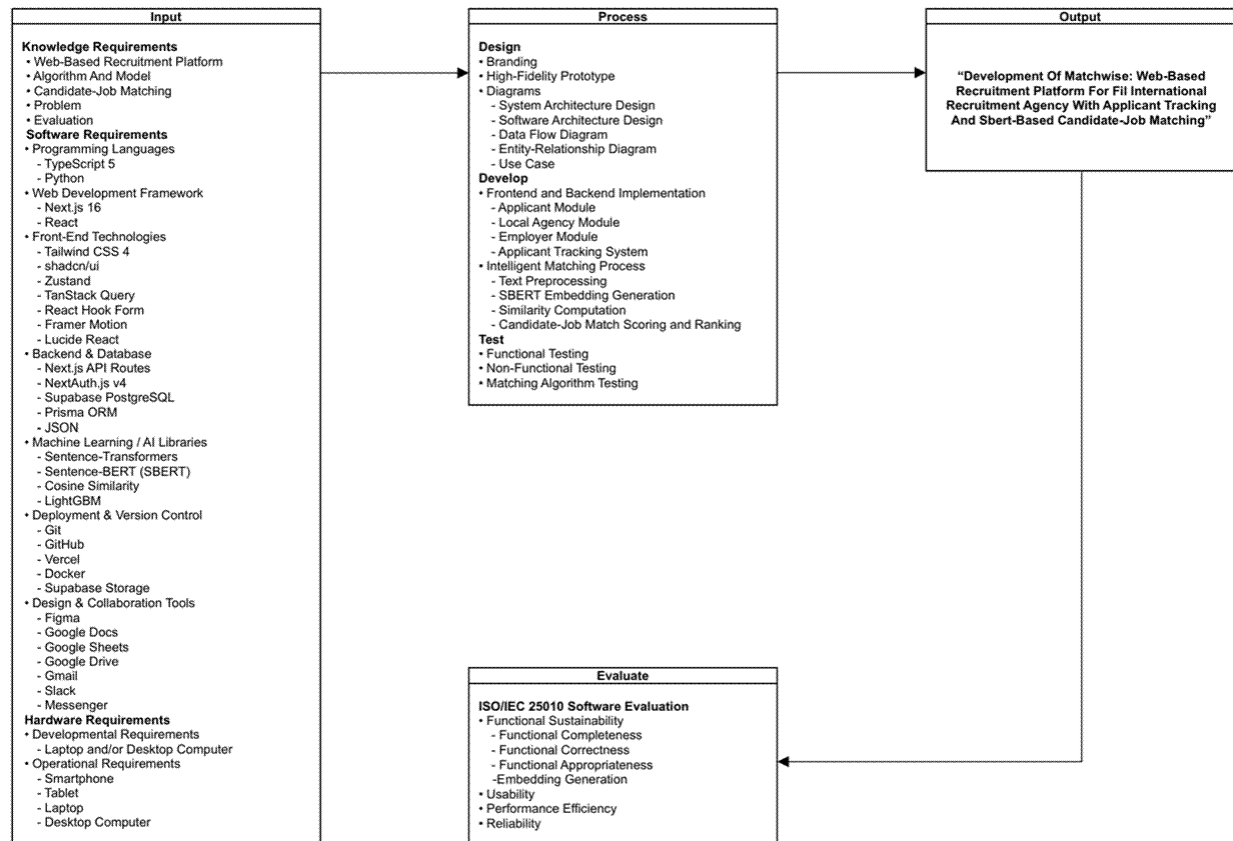


Figure 12. Conceptual Model of FIRA

The Input stage gathers the conceptual and technical resources that feed into the development of FIRA, organized into three sets: knowledge, software, and hardware requirements. Knowledge inputs include domain topics such as web-based recruitment platforms and applicant tracking systems, the algorithmic foundations of Sentence-BERT, cosine similarity, and LightGBM, candidate-job matching methodologies and their ethical concerns, the regulatory environment of Philippine overseas employment under RA 8042 and its amendment RA 10022, and the evaluation methods appropriate to intelligent recruitment systems. Software inputs list the practical tools needed to build the system, including collaboration and

documentation tools such as Google Docs, Google Sheets, Slack, and Messenger, UI/UX design through Figma, the core framework Next.js 16 with TypeScript running on the Bun runtime, frontend technologies including Tailwind CSS, shadcn/ui, Zustand, TanStack Query, React Hook Form, and Framer Motion, backend components including Next.js API Routes, NextAuth.js, and Prisma ORM, the machine learning stack comprising Python, Sentence-Transformers, and LightGBM, the database and storage through Supabase PostgreSQL and Supabase Storage, and version control through GitHub. Hardware inputs identify devices required for both development and operation, with developmental requirements covering laptops and desktops, and operational requirements covering smartphones, tablets, laptops, and desktops. Together, these inputs form the foundation that shapes technical choices, data collection, and implementation constraints throughout the study.

Building directly from these inputs, the Process phase applies an Agile cycle of design, development, and testing to turn requirements into working components. The Design stage produces project branding, a high-fidelity prototype, and core documentation including system architecture design, software architecture design, data flow diagrams, entity-relationship diagrams, and use cases that map how applicants, FIRA local agency staff, and foreign employers interact with the platform. Development implements the frontend and backend of three modules, namely the Applicant Module, the Local Agency Module, and the Employer Module, and embeds the algorithmic pipeline comprising SBERT for embedding generation, cosine similarity for similarity scoring, and LightGBM for candidate ranking. Testing runs across three areas: functional testing to verify feature correctness, non-functional testing to assess

performance and responsiveness, and algorithm testing to validate matching accuracy and analyze errors, ensuring that all features, interface components, and matching components work together as designed.

Those process activities yield the Output: a fully developed and deployable web-based recruitment platform titled FIRA: A Web-Based Recruitment Platform with Applicant Tracking and SBERT-Based Candidate-Job Matching for Fil International Recruitment Agency. The system provides applicant-facing features for account creation, job browsing, application submission, application status monitoring, and document management; local agency features for job posting, applicant screening, applicant endorsement, and recruitment reporting; and employer features for endorsed candidate review, selection, and offer management. The matching component produces ranked candidate lists with associated similarity scores and matched and missing skill identification, presented as decision-support information that supports rather than replaces human evaluation.

Finally, the Evaluate phase closes the loop by measuring software quality and matching accuracy, feeding results back into the development cycle. Evaluation applies ISO/IEC 25010 software quality criteria across functional suitability, usability, performance efficiency, and reliability, assessed by IT experts through a structured software evaluation questionnaire. Usability is separately evaluated by intended end users, including applicants, FIRA staff, and employers, through the System Usability Scale and the Technology Acceptance Model. Matching accuracy is evaluated using accuracy, precision, recall, F1 score, and information retrieval ranking metrics, with cross-validation and overfitting checks ensuring that the model generalizes to new

applicant-job pairs. Findings from these evaluations identify defects, usability gaps, and matching shortcomings that inform refinements, ensuring the platform meets both technical standards and the practical needs of its intended users.

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